SIEMENS Healthineers

MAGNETOM

Troubleshooting Guide
System

Patient Handling

Avanto
Espree

Document Version

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1.1 General

This section of the TSG deals with problems regarding the PMU. The strategies and procedures provide the necessary information for servicing the PMU. An advanced trouble shooting procedure is included in this document.

The primary components of the **PMU** Physiological Measurement Unit are:

PDAU-Physiological Data Acquisition unit

PERU-Physiological ECG and Respiratory Sensor Unit

PPU-Peripheral Pulse Unit

The physiological signals are shown on the PMU display at the magnet and on the physiological display UI.

The PERU and the PPU are supplied with a small black connector. This connector prevents the rechargeable battery from discharging. Disconnect the connectors if the units are in operation.

A new PDAU is introduced with software version VB13. Older software versions do not support the new PDAU, material no.10096522.

1.2 Additionally required documents

Function Description "Patient Handling" chapter

Replacement of Parts "Patient Handling" chapter

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1.3 Strategy

The primary tool for PMU troubleshooting is the **PMU** test, which is available in the Service Software platform under **Test Tools**. The error messages are shown in the event log. The following table is designed to provide the CSE with the error symptom and identify the specified FRU in the PMU system. **First** use the **Test Tools** before starting any other service step.

The "Replacement of Parts" folder lists the necessary adjustments and the final checks when replacing the FRU.

Tools

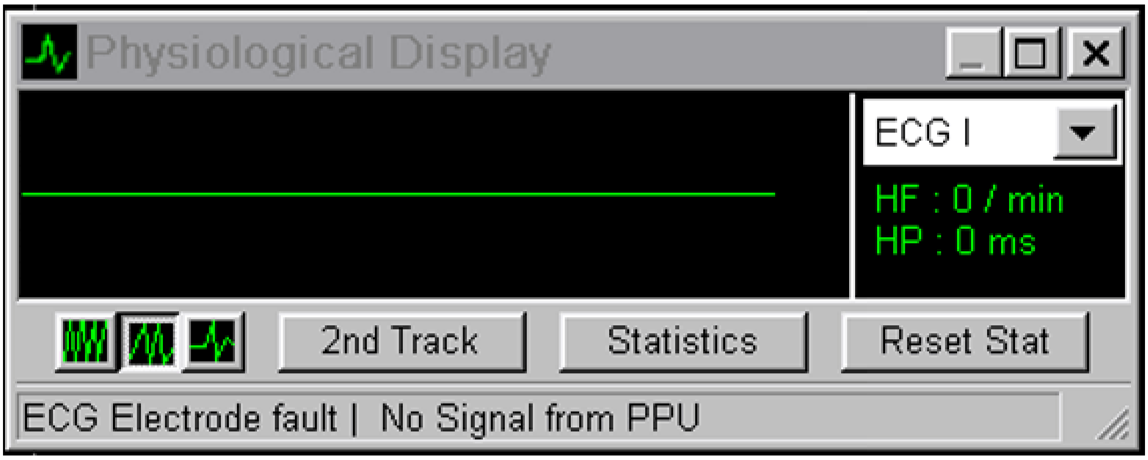
Non-magnetic tool kit

1.3.1 Chart 01

The error messages shown under **symptom** occur in the physiological display UI.

Messages in the physiological display user interface (UI) about the PMU status; see example.

Fig. 1: No signal on physiological display UI



Symptom	Area of Trouble (Components/function)	Probable Cause	Service action	Reference
01. No valid status from PDAU available	PDAU/LWL_INT D6 (AMC)/ Power supply cables	PDAU/LWL_INT D6 (AMC)/ Power supply cables	Run SeSo Testtools: **PMU Test**, Test type: Comm/Status and Interactive Test will fail; continue with next step	(→ Fig. 6 Page 13)
02. No valid status from PDAU available	PDAU	Fiber optic cable U1/U2 Fiber optic cable U6/U7/ D6 LWL_INT (AMC)	Check fiber optic cables Disconnect U1/U3 at PDAU and check if both LED's light up	n.a.
		PDAU power voltage missing	Remove cover of Ptab electronic. Check cable X6 (W4125). Check connector X64 at SUPF A4120 Ptab Meas. power voltage (approx. 16 Vdc)	**Replacement of Parts** covers “guide rails”
		PDAU defective	If LEDs U1/U3 do not light up, replace PDAU	**Replacement of Parts** “PDAU”
01. No valid status from PDAU available	LWL_INT D6 (AMC)	LWL_INT D6 (AMC), power supply, jumper settings	Check power voltage LWL_INT D6 (AMC) V2/V3 (green, +5/12V); if one of these voltages fails, V5 (red) will light up. Pull FOC U6/U7 at LWL_INT D6 and check if LED U6 lights up. Check FOC; to be connected to U7, LWL_INT D6 must light up. Check jumper setting on D6 (AMC) X42/43/44 1-2 If jumper and voltage are o.k., replace LWL_INT D6 (AMC)	(→ Fig. 9 Page 15)

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1.3.2 Chart 02

If no signal is present on the physio or PMU Display at the magnet, review additional troubleshooting hints.				
Symptom	**Area of Trouble (Components/function)**	**Probable Cause**	**Service action**	**Reference**
No ECG signal is displayed on the physio display or PMU display at the magnet	PERU	PERU, antenna, antenna cable, PDAU	Run SeSo Testtools: **PMU Test**, Test type:Comm/Status and Interactive (PERU positioned on table, electrode cables not connected to patient). Test o.k, test signal visible. (**Note** "ECG electrode fault" and "No signal from PPU) message in the phys. display is normal)	(→ Fig. 6 Page 13) (→ Fig. 3 Page 12) (→ Fig. 5 Page 13)
		Electrode cables defective	Use a screwdriver to establish a short connection between all electrode clips. If the red LED (electrode fault) still blinks, replace the PERU	(→ Fig. 8 Page 14) (→ Fig. 7 Page 14)
		A second PERU is located somewhere in the examination room.	Remove the spare PERU from the examination room and insert it into the sensor charger.	
		Application problems (e.g., dry electrodes)	Refer to system manual; contact application specialist	System manual
No respiratory signal is displayed on the physio display or PMU display at the magnet	PERU	PERU, antenna, antenna cable, PDAU, respiratory cushion	Apply the respiratory cushion to your abdomen (refer to system manual). Check if the signal is displayed on the physio or PMU display at the magnet.	You may need a second person for help
	PDAU	Respiratory cushion	Press the respiratory cushion with your fingers and check if there is air flow at the connector. If the previous tests are o.k. and you still have no respiratory signal, replace the PERU.	

If no signal is present on the physio or PMU Display at the magnet, review additional troubleshooting hints.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service action	Reference
No pulse signal displayed in the physio display or PMU display at the magnet	PPU	PPU, antenna, antenna cable, PDAU	Run SeSo Testtools: **PMU Test**, Test type: Comm/Status and Interactive (PERU positioned on table, electrode cables not connected to patient). Test o.k, test signal visible. (**Note** "ECG electrode fault" and "No signal from PPU) message in the phys. display is normal)	(→ Fig. 6 Page 13) (→ Fig. 3 Page 12)
		PPU	Position the PPU on the patient table. Attach the pulse sensor to your finger; check if the red LED (application fault) at the PPU stops flashing. Check if the pulse signal is displayed in the phys. display. If not o.k., replace the PPU.	
		A second PPU is located somewhere in the examination room.	Remove the spare PPU from the examination room and insert it into the sensor charger.	
		Application problems	Refer to system manual; contact application specialist	System manual

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1.3.3 Chart 03

PMU display in examination room				
Symptom	Area of Trouble (Components/function)	Probable Cause	Service action	Reference
PMU display dark	PMU display	PMU display, power supply, cables, PDAU	Check power cable (PDAU X5 to PMU display X1). Power voltage for PMU display ok? Power voltage for PDAU ok?	**Replacement of Parts** "PDAU"
No signal traces on PMU display	PMU display	PMU display, cables, PDAU	Check FOC U3/U4 PMU display/PDAU (Hints: LED U3/U4 on the PMU display are always dark. After reconnecting FOC's, a power ON reset of the display is necessary). Perform PDAU test; see previous steps. If all o.k., replace PMU display	
PMU display malfunctions, e.g. touch screen not working	PMU display	PMU display	After switching ON the display (display will stay dark for approx. 30 sec.), the test menu pushbutton appears. Press this button and perform the display test. If it is not functioning, replace the PMU display.	

Fig. 2: No signal on physiological display UI

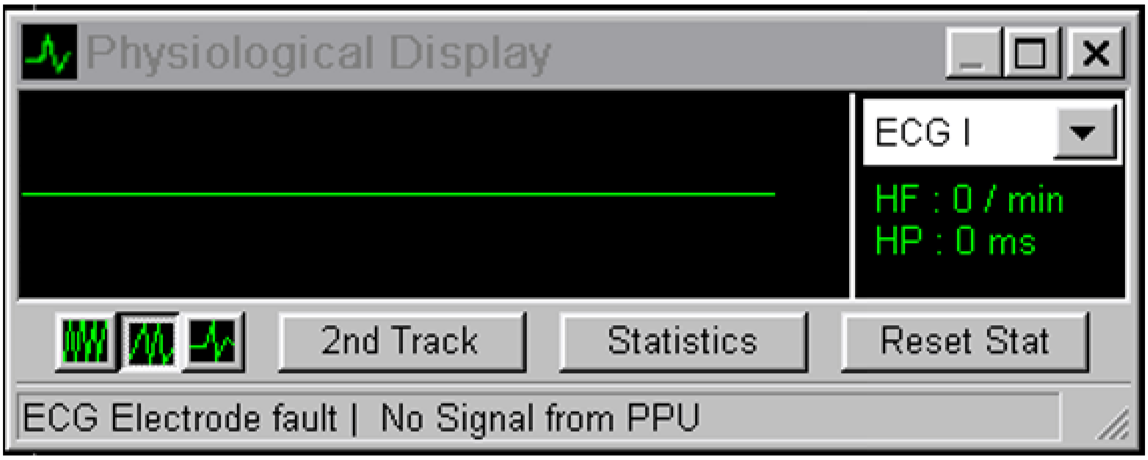

Fig. 3: PERU test signal with old revision level

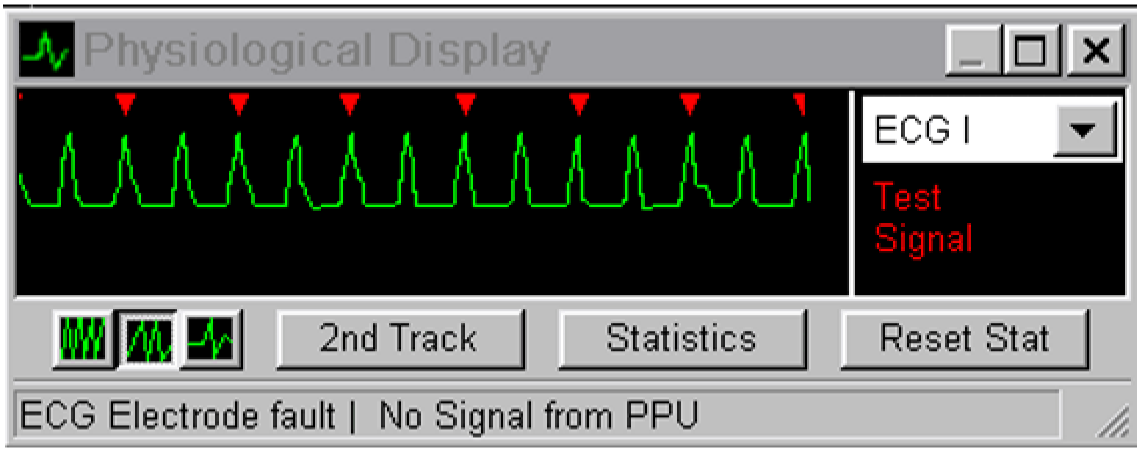

Fig. 4: PERU test signal newer revision level

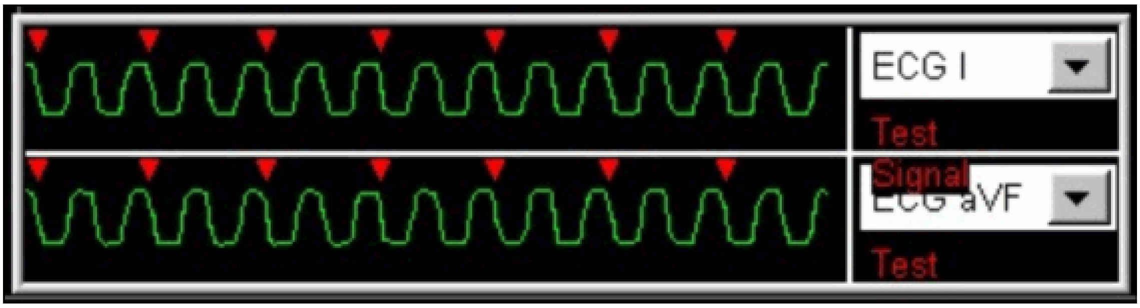

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Fig. 5: Help text on physiological display UI

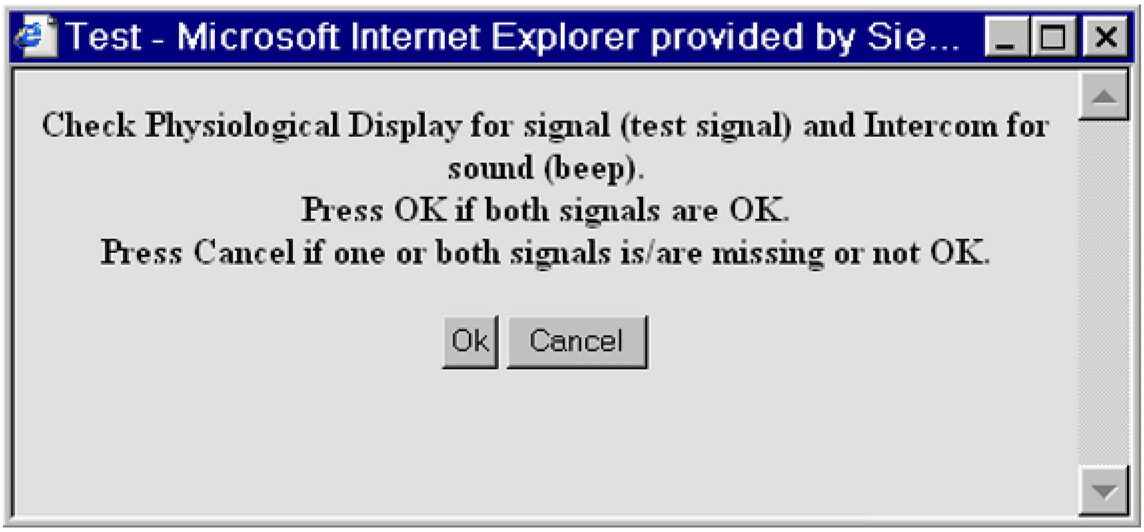


Fig. 6: Help text for PMU interactive test

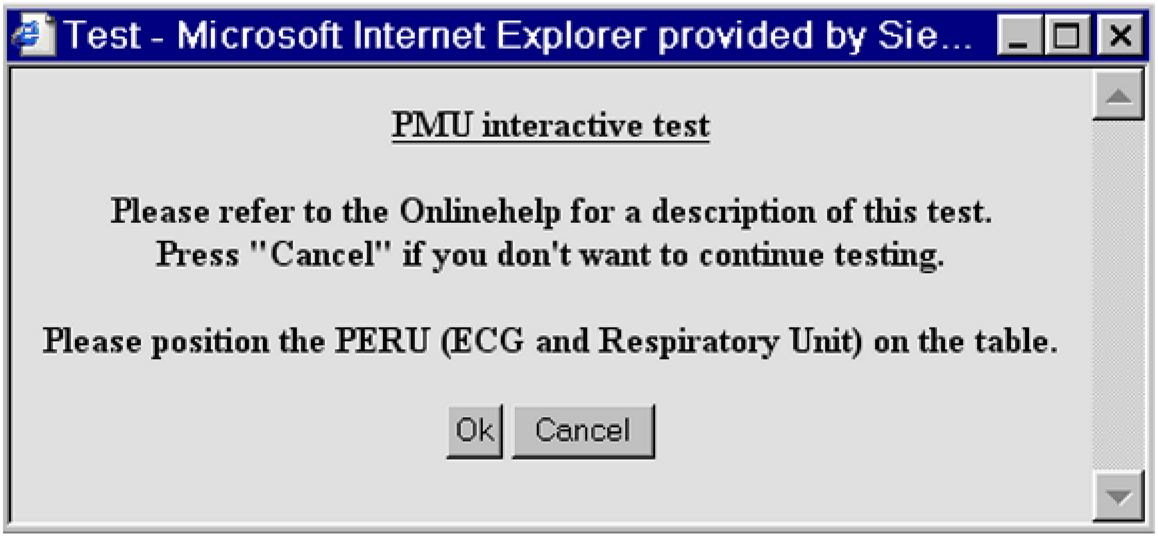
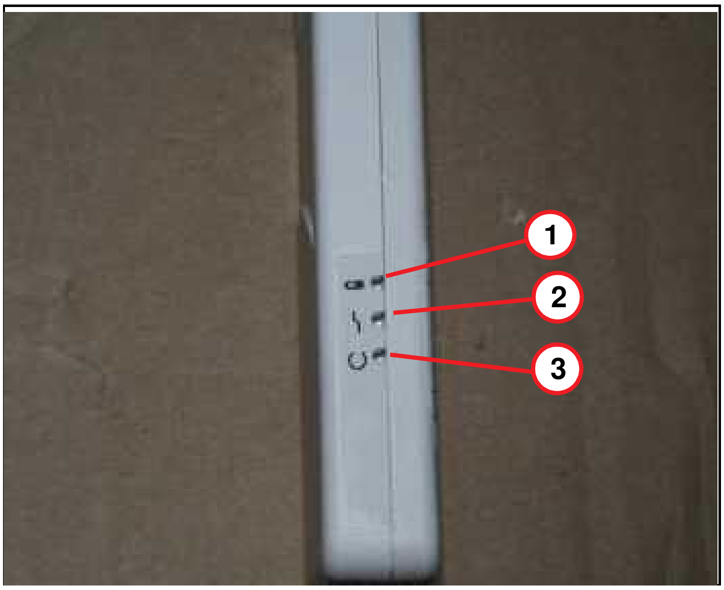


Fig. 7: LED indication



(1) Battery status (yellow)
(2) Application error, ecg or pulse sensor (red)
(3) Charge process (green)

Fig. 8: Short-circuited PERU electrode ports

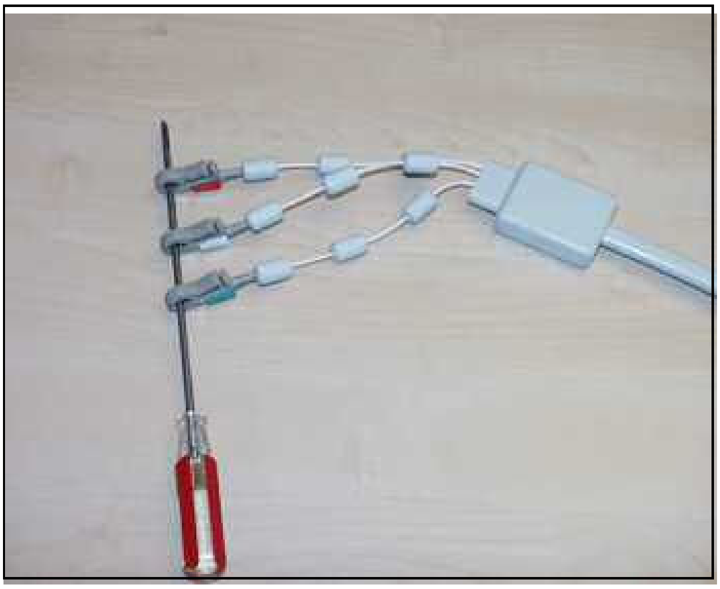


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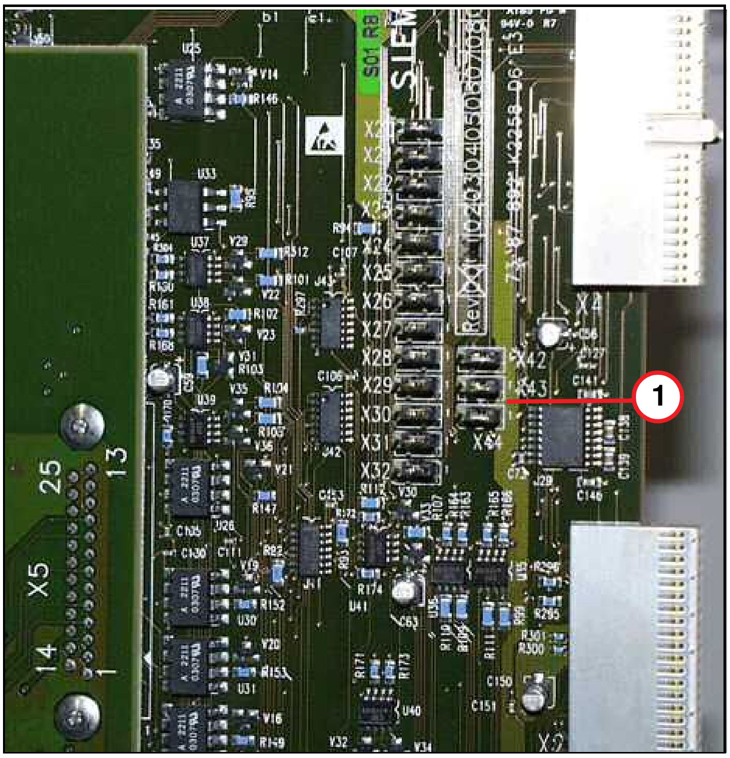
Fig. 9: Jumper setting D6 AMC



(1) Jumper setting on D6 AMC X42-X44 **1-2**

1.4 Advanced trouble shooting procedure for VCG triggering

Valid for software versions VB13A/VB15x/VC11A			
Symptom	**Possible root cause**	**Probable cause**	**Reference**
Zero line at the PhysioDisplay and the PMU Display	One or more electrodes do not form appropriate contact with the patient's skin - "ECG Electrode fault" is reported in the status line of the PhysioDisplay - see (➔ Fig. 10 Page 18)	Dry electrodes	Refer to (➔ Symptom: Zero line at the PhysioDisplay and the PMU Display / Page 18)
		Incorrect electrode type	Refer to (➔ Symptom: Zero line at the PhysioDisplay and the PMU Display / Page 18)
	Interference by a second PERU (➔ Fig. 11 Page 19)	another unused PERU	Refer to (➔ Symptom: Zero line at the PhysioDisplay and the PMU Display / Page 18)
	No signal received by PERU - "No Signal from PERU" is reported in the status line of the PhysioDisplay - see (➔ Fig. 12 Page 19)	The patient has **placed his hand on the transmitter box**	Refer to (➔ Symptom: Zero line at the PhysioDisplay and the PMU Display / Page 18)
		The **battery is low**	Refer to (➔ Symptom: Zero line at the PhysioDisplay and the PMU Display / Page 18)
Intermittent zero line at the PhysioDisplay and at the PMU Display	Interference by a second PERU	another unused PERU	Refer to (➔ Symptom: Intermittent zero line at the PhysioDisplay and at the PMU Display / Page 20)

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Valid for software versions VB13A/VB15x/VC11A			
Symptom	Possible root cause	Probable cause	Reference
Zero line at the PhysioDisplay, but correct signal at the PMU Display in the Examination Room	"No valid status from PDAU available" is reported in the status line of the PhysioDisplay, see - (→ Fig. 14 Page 21)	Potentially defective cables or PCB boards	Refer to (→ Symptom: Zero line at the PhysioDisplay, but correct signal at the PMU Display in the Examination Room / Page 21)
Triggering unreliable/ no triggering	Refer to (→ Symptom: Triggering unreliable/ no triggering / Page 21)	(→ Symptom: Triggering unreliable/ no triggering / Page 21)	(→ Symptom: Triggering unreliable/ no triggering / Page 21)

1.4.1 Symptom: Zero line at the PhysioDisplay and the PMU Display

1.4.1.1 Possible root causes

- One or more electrodes do not form appropriate contact with the patient's skin - "ECG Electrode fault" is reported in the status line of the PhysioDisplay - see (→ Fig. 10 Page 18)

Fig. 10: "Status line of PhysioDisplay" screenshot

  - **Due to dry electrodes** - please check the expiry date
  - **Due to an incorrect electrode type** - we highly recommend the Conmed 2700 Cleartrace electrodes, as they ensure reliable skin contact even for very time-consuming examinations. Other electrodes may tend to loose contact over time, especially for stress examinations, where the patients may tend to sweat. Due to **improper preparation** of the patient's **skin**: prepare the skin by means of NuPrep Skin Prepping Gel (please avoid alcohol, which removes the electrolytes from the skin), shave the patient's skin in the areas intended for the electrode placement (if necessary).

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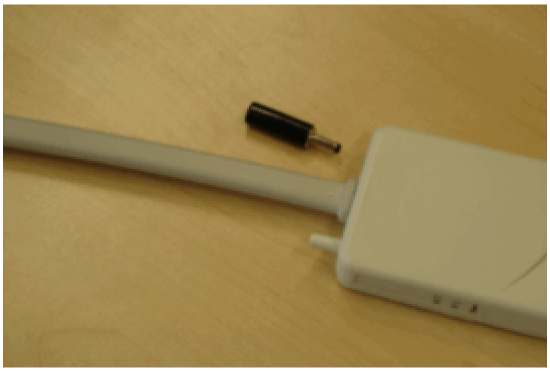
- **Interference by a second PERU**

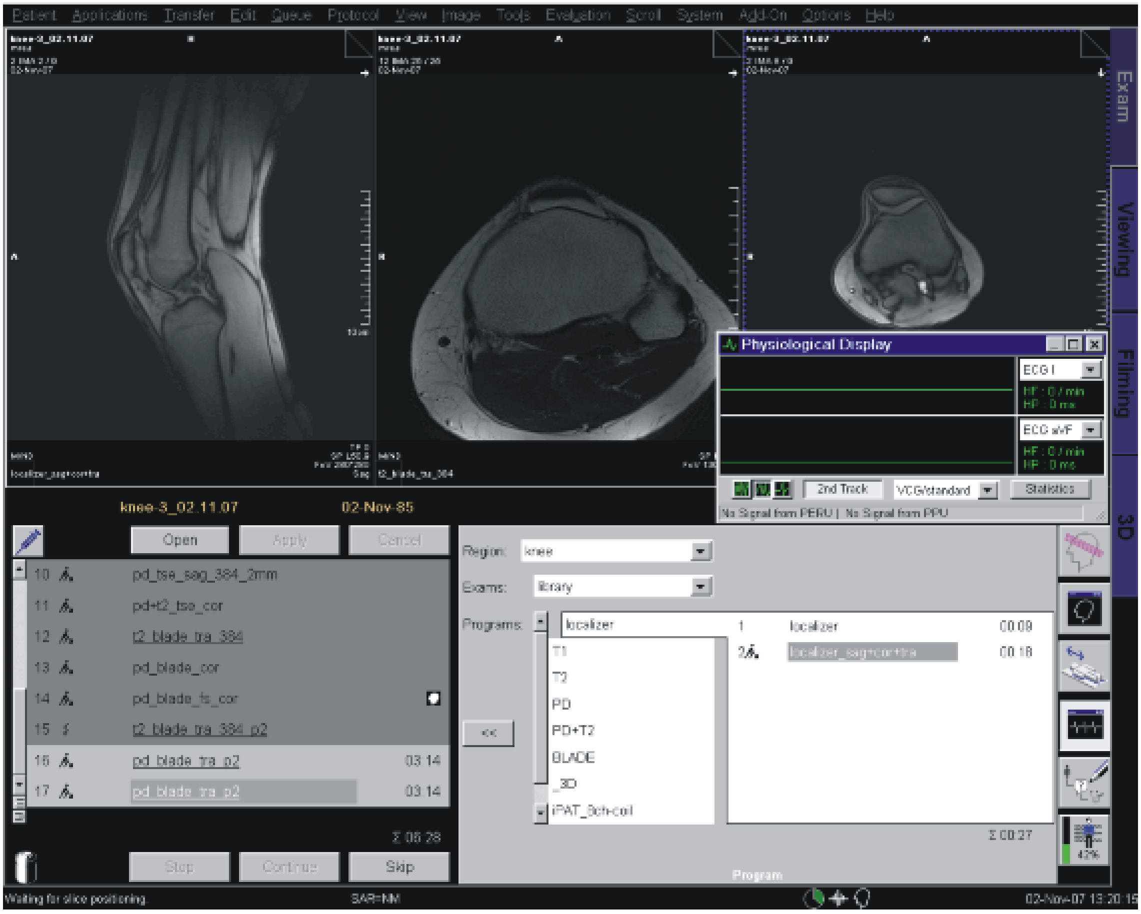
Fig. 11: Buckler of PERU



There is **another (unused) PERU** in the examination room or outside of it with the door open (e.g. not properly inserted into the charger), **which reports an ECG electrode fault** - see (➔ Fig. 10 Page 18). Place all unused PERUs in their chargers or plug in the buckler - see (➔ Fig. 11 Page 19).

- No signal received by PERU - "No Signal from PERU" is reported in the status line of the PhysioDisplay - see (➔ Fig. 12 Page 19)

Fig. 12: "No signal from PERU" screenshot



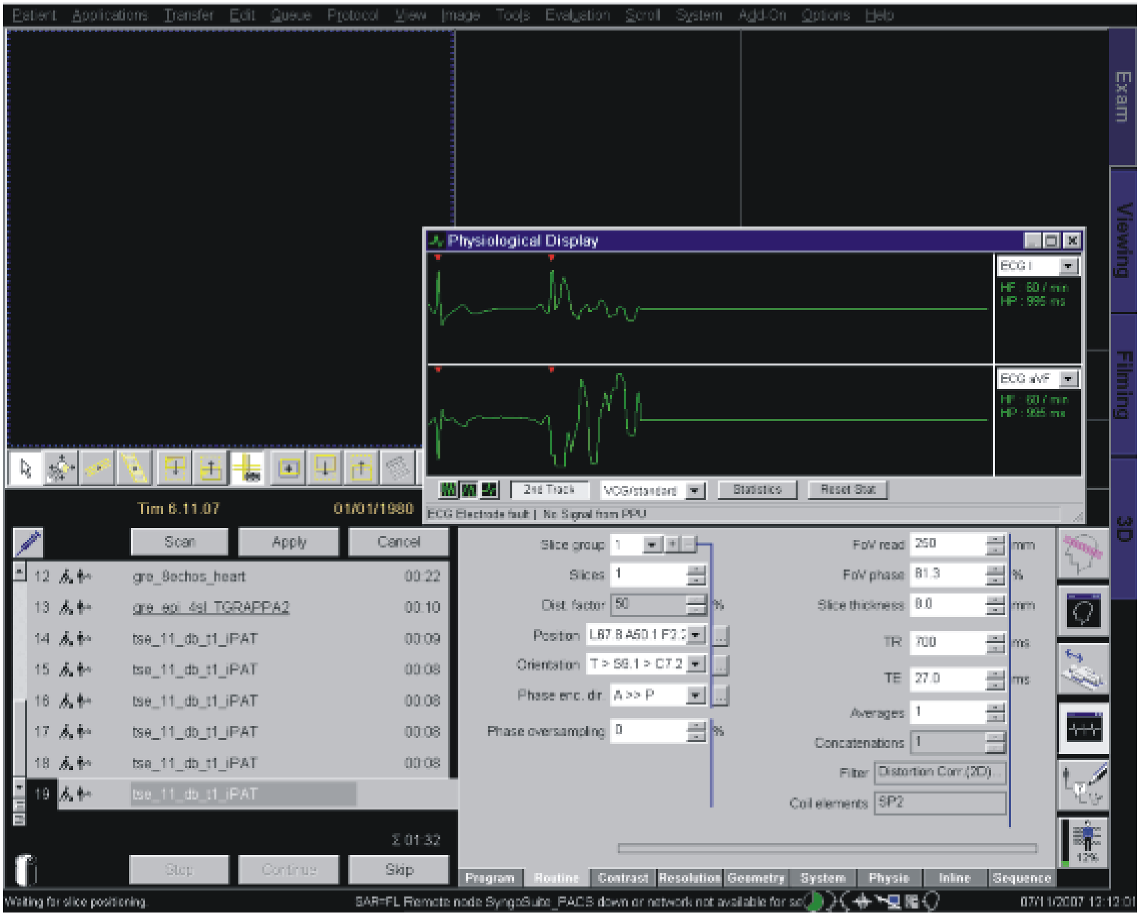
- The patient has **placed his hand on the transmitter box** (the larger of the two boxes), which leads to an excessive signal attenuation.
- The **battery is low** (none of the LEDs at the unit are flashing - prior to this behavior, the red LED starts flashing and a weak battery is reported in the status line of the

PhysioDisplay - you then have at least 1 more hour of operation before the unit will need to be recharged).

1.4.2 Symptom: Intermittent zero line at the PhysioDisplay and at the PMU Display

Intermittent zero lines with an "ECG Electrode fault" reported in the status line of the Physio Display - see (→ Fig. 13 Page 20):

Fig. 13: "ECG Electrode fault" screenshot



1.4.2.1 Possible root cause

- **Interference by a second PERU**

  There is **another (unused) PERU** in the examination room or outside of it with the door open, **which reports the "ECG electrode fault"**. Please place all unused PERUs accurately into their chargers (without any **flashing** green and red LEDs, just the yellow LED is allowed to be **illuminated** or all 3 LEDs should be dark, if the PERU has been fully charged).

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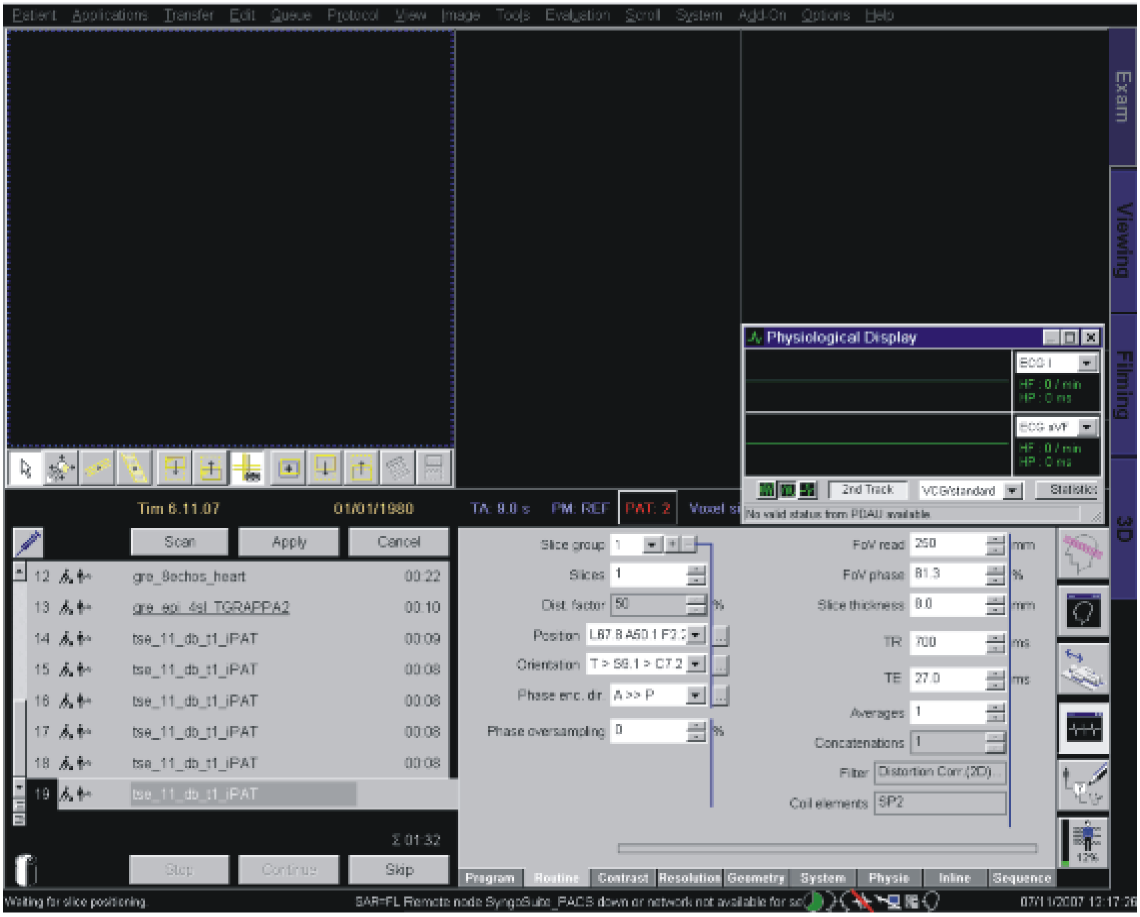
1.4.3 Symptom: Zero line at the PhysioDisplay, but correct signal at the PMU Display in the Examination Room

"No valid status from PDAU available" is reported in the status line of the PhysioDisplay, see - (➔ Fig. 14 Page 21)

Fig. 14: "No valid status from PDAU available" screenshot



Do not replace the PERU - it is fine in this case.

1.4.3.1 Possible root causes

Potentially defective cables or PCB boards.

1.4.4 Symptom: Triggering unreliable/ no triggering

In our experience most of the problems with VCG gating relate to disadvantageous patient handling during the learning phase, which can be quite easily improved for most of the examinations.

Please find below the most important notes, **valid for the software versions VB13A, VB15A, VB15B and VC11A:**

1.4.4.1 Application note for stable VCG triggering

To ensure reliable operation of the VCG triggering - especially for B13 customers

- it is absolutely essential, that the following notes are observed.
- After attaching the PERU as described in the System Manual, the two leads (aVF and I) are evaluated. The R-waves should be clearly recognizable for both leads and as a result suitable for triggering during the examination.
- After positioning the Body Matrix, wait at least 10 heart beats for the learning phase to finish before moving the patient into the magnet (in table home or near home position).

1.4.4.2 Detailed information

1. The learning phase of the VCG is complete, when the patient table is located within 300 mm (118 inches) from the home position (learning range). The home position works best due to the lowest B0 stray field, which ensures T waves are minimally elevated by the magnetohydrodynamic effect.
2. The learning phase is interrupted as soon as the table is moved.
3. If the table is stopped within the learning range, the learning phase is continued.
4. At least 10 regular heartbeats are needed for the learning phase to finish successfully (regular means 10 heartbeats acquired with the patient lying still **and** the patient table remains at the same position within the learning range). After a first set of 10 **regular** heartbeats has been learned, another learning period of 10 heartbeats is started. If a second set of 10 regular heartbeats has been acquired before the table starts moving, this second set will be taken instead of the first one (and so on). This should ensure that at least 10 regular heartbeats have been learned at the same table position.
5. If the table is moved more than 300 mm away from home position, the learning phase is terminated.
6. Another (new) learning phase is started only if the patient is moved out of the bore to a position within the learning range.
7. If the PERU is connected to the attached electrodes **after** the patient table has been already moved into the magnet **beyond the learning range**, no triggers can be generated by the VCG algorithm due to the lack of learning data and therefore the **red**

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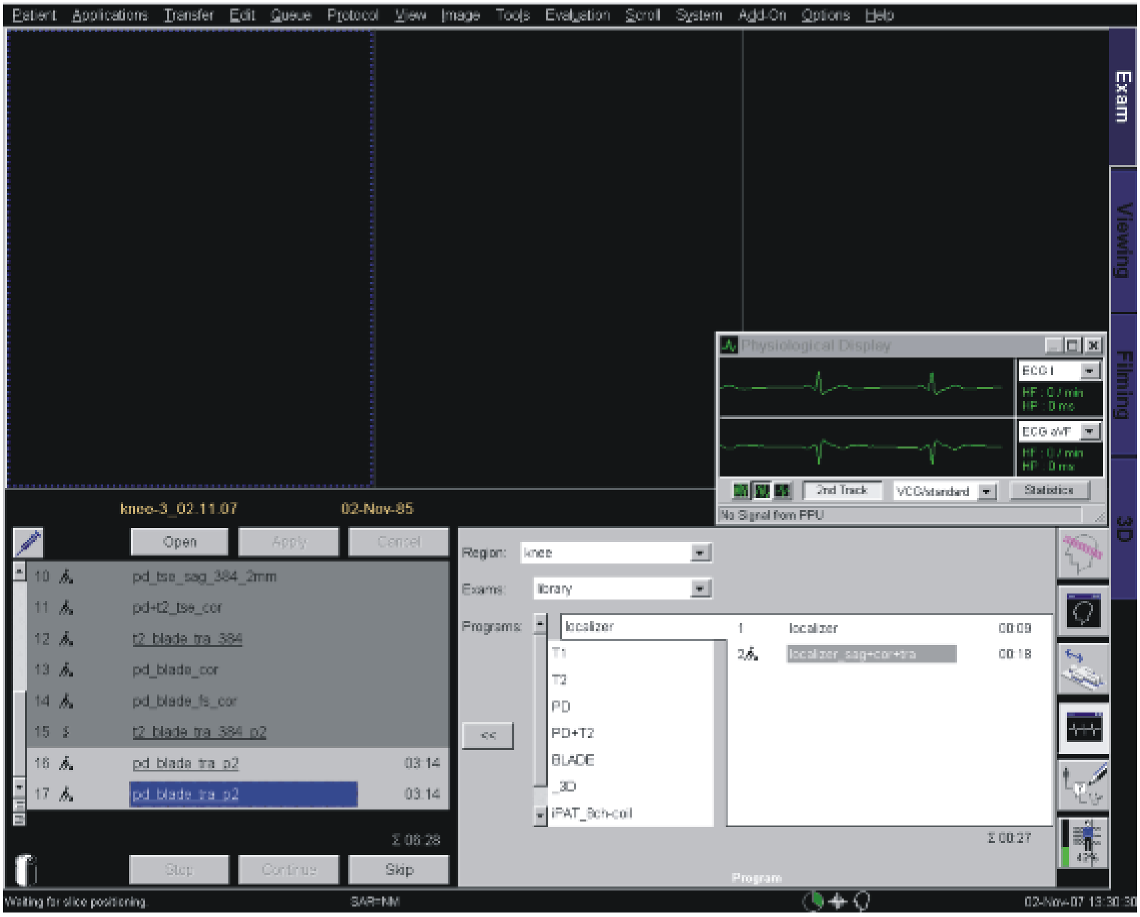
trigger triangles in the Physio Display are **missing** - see (→ Fig. 15 Page 23) and **"waiting for trigger pulse"** is reported, when a protocol is run:

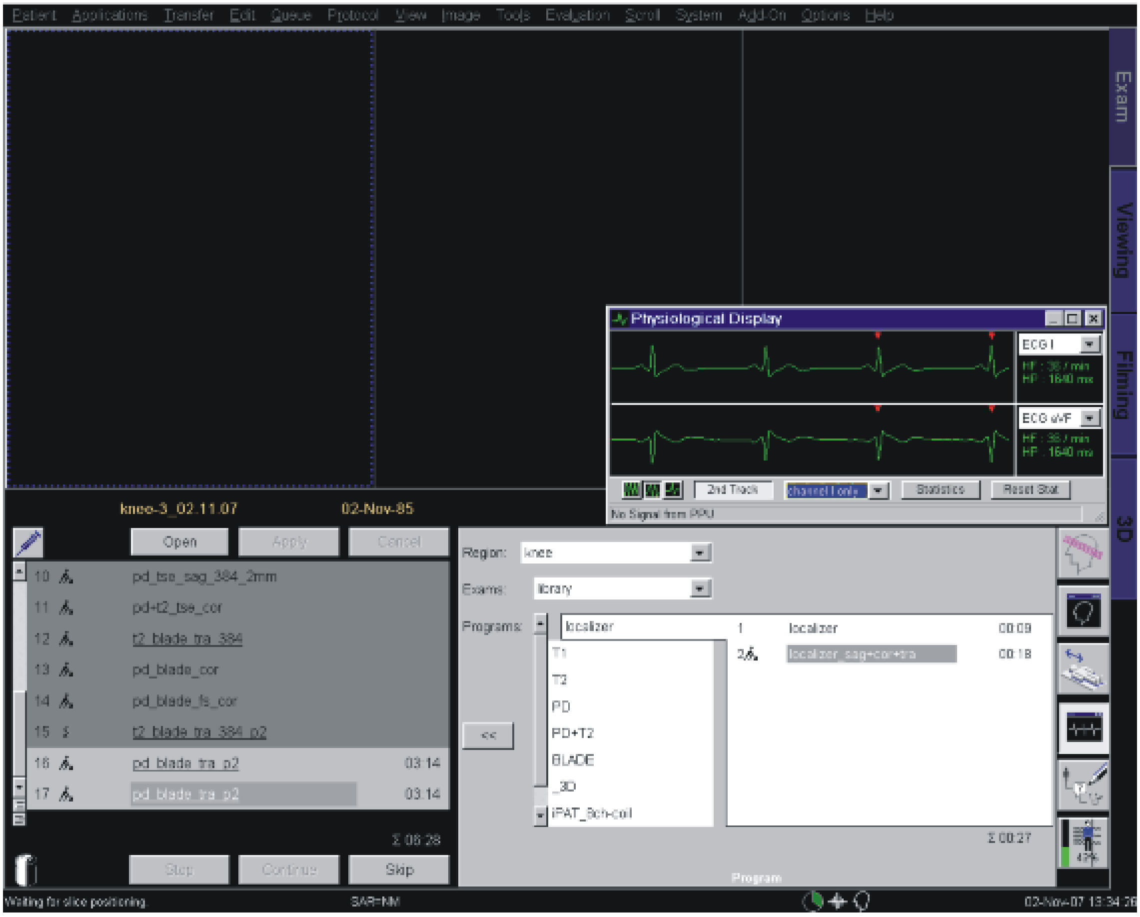
Fig. 15: "Waiting for trigger pulse" screenshot



Moving the Patient Table out to the home position solves the problem because the learning phase is then initially started.

If it is too disadvantageous to move the patient table back to the home position, **switching to** the triggering modes "channel I only" or "channel AVF only" at the PhysioDisplay circumvents this problem, as those two modes don't use the learning data (please observe the notes regarding the **one channel triggering** modes below in this special case). See (➔ Fig. 16 Page 24):

Fig. 16: "Switching to triggering modes" screenshot



8. In the default trigger mode **"standard_VCG"** both ECG channels are used for triggering. It is highly recommended to **commonly use this mode.** It also works fine for most of the examinations. If one of the ECG channels is significantly smaller than the other one,

> a display of both ECG channels in the PhysioDisplay simplifies this assessment.

If no R waves are visible in one of the channels (and less than 1 star ("*") is indicated at the **Patient Table Display** for VB15A and VB15B/ VC11A), it is recommended to use the other channel for triggering ("channel I only" and "channel aVF only" modes), or try to optimize the electrode placement to gain a more distinct R wave in the weaker channel.

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1.4.5 Improvements with VB15A, VB15B

- As a further improvement, VB15A/ VB15B feature a display of the ECG amplitudes at the Patient Table Display - see (→ Fig. 17 Page 25)

Fig. 17: Patient table display for VB15A/VB15B

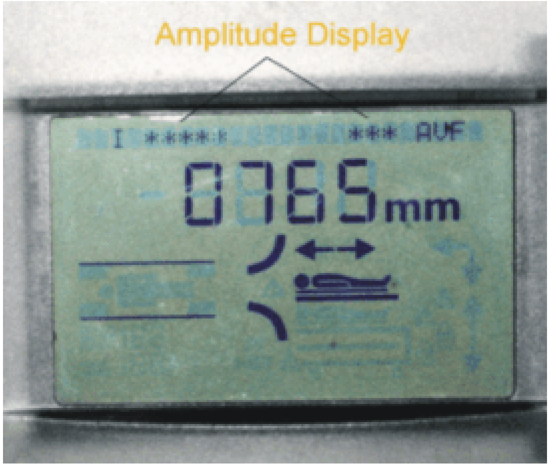
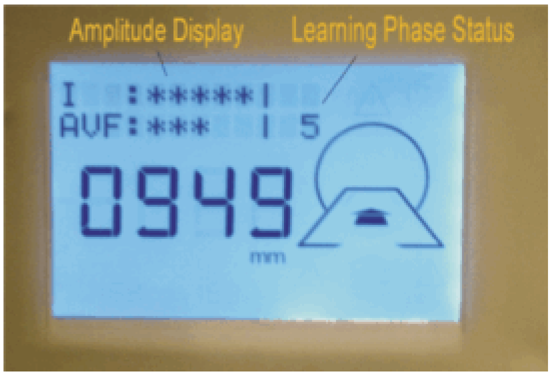


Fig. 18: Patient table display for VC11A



1.4.5.1 Working with the Amplitude Display (all software versions indicated above)

Ensure that at least 2 stars ("*") are shown on one ECG channel (I or AVF) of the table display. The other channel must show at least 1 star.

In the example above, 5 stars are shown for channel I and 3 stars for channel AVF, which means that the amplitude of both channels should be sufficient for reliable triggering.

Depending on the electrode position, there can be a very strongly elevated T-wave particularly for 3 T systems (caused by the magnetohydrodynamic effect), which may still lead to trigger problems when the patient has been moved into the bore. Triggering on "channel I only" or "channel AVF only" (whichever has the lower elevated T-Wave) can potentially solve the trigger problems. Another possibility is repositioning the electrodes, e.g. moving the "white" electrode away from the aorta and keep on working with "standard VCG" triggering.

1.4.6 Improvements with VC11A

- VC11A also now features a display of the learning phase status at the Patient Table Display:

Fig. 19: Learning phase status

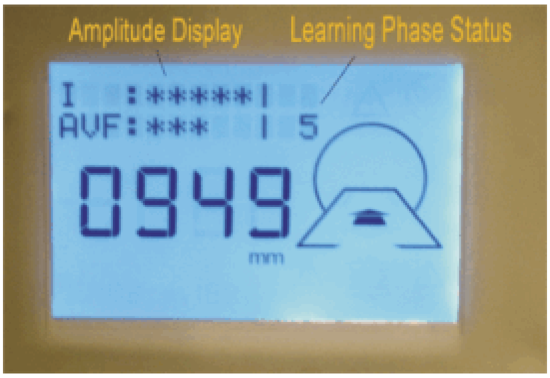


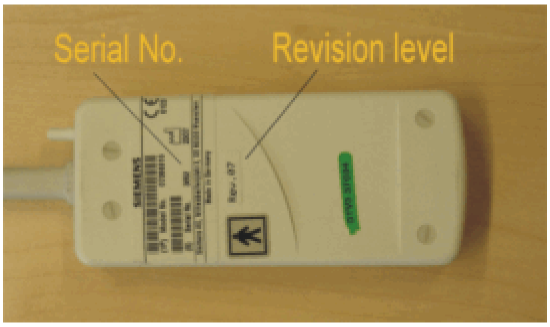
Fig. 20: Learning phase successful

1.4.6.1 Working with the learning phase status display:

The progress of the learning phase is also displayed on the table display - see (➔ Fig. 19 Page 26). An active learning phase is indicated by the number of heartbeats. The system continuously counts up from 1 to 10 until the patient table is moved into the magnet. The number 10 indicates that the learning phase has been finished successfully - see (➔ Fig. 20 Page 26).

1.4.7 Problems that cannot be resolved with the above information

Fig. 21: Serial number, revision level



If all mentioned tips do not resolve a VCG triggering problem, it would be very helpful if you could provide us with the serial number and the hardware revision level of the PERU - see (➔ Fig. 21 Page 26).

In addition to this, it helps us to diagnose further problems if you could send us ECG log files (please refer to the instructions below for the different software versions).

1.4.7.1 Recording of physiological signals with VB11x, VB13A, VB15A and VB15B

- The resulting log files can be found in the directory c:\MedCom\log

  The extension of the ECG files is .ecg
- Necessary inputs are indicated in italics

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Open Windows XP Start Menu:

Menu item "Run..."

Open: telnet mpcu

The VxWorks shell is entered...

Start/ Stop the signal logging (e.g. for the ECG signal):

```
-> fMenu
+++++++++++++++++++++++++++++++++++++++++++ Menu V1.2 ++++
+ / MPCU Application Menu
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
+
+ 1 - BGT-MMC Pruefung...
+ 2 - PCI Karte/Treiber ...
+ 3 -  Kommunikation zu den DSPs...
+ 4 - PMU ...
+ 5 - Trace Utilities ...
+ 6 - VxWorks System Info ...
+
++++++++++++++++++++++++++++++++++++++++++++++ h=Help ++++
Select number [1-6]: 4

+++++++++++++++++++++++++++++++++++++++++++ Menu V1.2 ++++
+ /4/ PMU
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
+
+ 1 -  Kommandos zum Frontend...
+ 2 -  Debugausgaben steuern...
+ 3 - Signal Logging ...
+
++++++++++++++++++++++++++++++++++++++++++++++ h=Help ++++
Select number [1-3]: 3

+++++++++++++++++++++++++++++++++++++++++++ Menu V1.2 ++++
+ /4/3/ Signal Logging
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
+
+ 1 - start ECG
```

```
+ 2 - stop ECG
+ 3 - start RESP
+ 4 - stop RESP
+ 5 - start PULS
+ 6 - stop PULS
+ 7 - start EXT
+ 8 - stop EXT
+ 9 - start all signals
+ 10 - stop all signals
+
++++++++++++++++++++++++++++++++++++++++++++++ h=Help ++++
Select number [1-10]: 1

Enter log file name (without path and extension): tfi_perf_220208
```

Please remember to stop the logging after the session:

```
+++++++++++++++++++++++++++++++++++++++++++ Menu V1.2 ++++
+ /4/3/ Signal Logging
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
+
+ 1 - start ECG
+ 2 - stop ECG
+ 3 - start RESP
+ 4 - stop RESP
+ 5 - start PULS
+ 6 - stop PULS
+ 7 - start EXT
+ 8 - stop EXT
+ 9 - start all signals
+ 10 - stop all signals
+
++++++++++++++++++++++++++++++++++++++++++++++ h=Help ++++
Select number [1-10]: 2

+++++++++++++++++++++++++++++++++++++++++++ Menu V1.2 ++++
+ /4/3/ Signal Logging
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
```


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```
+
+ 1 - start ECG
+ 2 - stop ECG
+ 3 - start RESP
+ 4 - stop RESP
+ 5 - start PULS
+ 6 - stop PULS
+ 7 - start EXT
+ 8 - stop EXT
+ 9 - start all signals
+ 10 - stop all signals
+
+++++++++++++++++++++++++++++++++++++++++++++ h=Help ++++
Select number [1-10]:
```
ex (leave the test menu)

After the logging session: Please close the telnet session:

-> logout

> Within the test menu, you can go up one level by typing . . (two full stops)

1.4.7.2 Recording of physiological signals with VC11A

- The resulting log files can be found in the directory c:\MedCom\log
  The extension of the ECG files is .ecg
- Necessary inputs are indicated in italics

Open Windows XP Start Menu:

Menu item "Run..."

Open: ideacmdtool

The ideacmdtool menu is entered...

Start/ Stop the signal logging (e.g. for the ECG signal):

```
--------------------------------------------------
IDEACmdTool Menu
--------------------------------------------------
1. PMU control...
2. PAS unload
3. Measure
```

```
4. Image target
5. Switches...
6. Set Default
--------------------------------------------------
Please enter number or unique text of a cmd. ('q' for quit)
--------------------------------------------------
> 1

--------------------------------------------------
PMU control Menu
--------------------------------------------------
1.PMU Signal simulation
2.PMU Signal logging
--------------------------------------------------
Please enter number or unique text of a cmd. ('q' for quit)
--------------------------------------------------
PMU> 2

--------------------------------------------------
Menu
--------------------------------------------------
1. dump
2. startHostLogECG
3. stopHostLogECG
4. startHostLogRespiration
5. stopHostLogRespiration
6. startHostLogPulse
7. stopHostLogPulse
8. startHostLogExternal
9. stopHostLogExternal
10. startHostLogAll
11. stopHostLogAll
--------------------------------------------------
Please enter number or unique text of a cmd. ('q' for quit)
--------------------------------------------------
Cmd > 2
enter filename: fl_trig140208
```


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```
command startHostLogECG was executed successfully
```

Please remember to stop the logging after the session:

```
--------------------------------------------------
Menu
--------------------------------------------------
1. dump
2. startHostLogECG
3. stopHostLogECG
4. startHostLogRespiration
5. stopHostLogRespiration
6. startHostLogPulse
7. stopHostLogPulse
8. startHostLogExternal
9. stopHostLogExternal
10. startHostLogAll
11. stopHostLogAll
--------------------------------------------------
Please enter number or unique text of a cmd. ('q' for quit)
--------------------------------------------------
Cmd > 3

command stopHostLogECG was executed successfully

--------------------------------------------------
Menu
--------------------------------------------------
1. dump
2. startHostLogECG
3. stopHostLogECG
4. startHostLogRespiration
5. stopHostLogRespiration
6. startHostLogPulse
7. stopHostLogPulse
8. startHostLogExternal
9. stopHostLogExternal
10. startHostLogAll
```

```
11. stopHostLogAll
--------------------------------------------------
Please enter number or unique text of a cmd. ('q' for quit)
--------------------------------------------------
Cmd > q

--------------------------------------------------
PMU control Menu
--------------------------------------------------
1.PMU Signal simulation
2.PMU signal logging
--------------------------------------------------
Please enter number or unique text of a cmd. ('q' for quit)
--------------------------------------------------
PMU> q

--------------------------------------------------
IDEACmdTool Menu
--------------------------------------------------
1. PMU control...
2. PAS unload
3. Measure
4. Image target
5. Switches...
6. Set Default
--------------------------------------------------
Please enter number or unique text of a cmd. ('q' for quit)
--------------------------------------------------
> q (ideacmdtool exits)
```


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2.1 General

This section of the TSG deals with problems regarding the patient table and the trolley (RTT **R**emovable **T**able **T**op). The strategies and procedures provide the necessary information to maintain the patient table.

Please add the error ID, the error message, and a detailed description of the problem (e.g. position of table, load, error frequency...) to any replaced PCB.

2.2 Strategy

The main tool for patient table troubleshooting are the **error messages** delivered from the patient table. The **error messages** are shown in the error log file from the local service platform. A detailed description about the error is part of the error message. All message IDs are listed in the Healthcare Services Knowledge Base (SKB). Each error message includes the **error ID, message text, explanation and action**. The patient table communication and status test are also available in the service software platform under "**Test Tools**" and in the service software for the patient table. The patient table itself comes with additional service functions, such as the **service mode**, **display of the current errors**, **debug display** and **debug marking**. To supplement the error messages, additional tables provide the CSE with information on troubleshooting **switches, sensors, jumpers, and LEDs**. FRU replacements are described in the REP patient handling chapter.

For an explanation of the basic functions of the patient table, please refer to the **Function Description** in the patient handling chapter.

2.2.1 Display of current errors

The table control is able to report errors in the following ways:

- Via CAN bus to the host: Here, the error codes are identified and connected to the corresponding error text stored in syngo MR.
- Numerically on the patient table display: ("Error: nn") If there are multiple errors, the display cycles through them sequentially. The number displayed corresponds to the error message ID in the event log and TSG.
- Error text at the patient table display: If there has been no CAN-host communication since the last power-up, the error text is displayed in the text lines. (Several errors: display sequentially)

2.2.2 Additionally required documents

Function Description "Patient Handling" chapter

Replacement of Parts "Patient Handling" chapter

Error messages Patient Handling → Healthcare Services Knowledge Base (SKB)

2.3 Service mode

2.3.1 General

Components of the table hardware can be checked using the service menu:

- switches and sensors
- Control panels (including brightness adjustment)
- displays (including contrast adjustment)

2.3.2 Activating the service mode

On the control panel, simultaneously press the buttons "**Air-", "Air+" and "Light+".** Service mode starts once the buttons are released.

> During active service mode, the normal table functions (motor-driven movements) are not available

2.3.3 Operating the service mode

The function selected is displayed in the upper left of the table display (exception: display test during automatic operation)

It is possible to switch between individual functions in service mode using the “HDP+” and “HDP-” keys.

“**HDP+**” increases the function number (up to 9), and “**HDP-**” decreases it by one.

The following functions are available:

- Function 1: Loadware version, absolute 1st position
- Function 2: LCD test and contrast adjustment
- Function 3: Keyboard test
- Function 4: LED test and brightness adjustment
- **Very useful** Function 5: Switch test
- Function 6: Temperature display

Function 9 is used to exit the service mode (press “Start/Stop” to confirm).

2.3.3.1 Function 1: Loadware version, absolute 1st position

The current version of the table loadware is displayed after the function number (ticker). Example: "PTABAppl_VA04I jan 22 2004 [PT005 0000.0004-0443 22.01.2004]"

The 4-digit position display under the text line indicates the absolute table position in mm (based on reference position) in the operating mode.

2.3.3.2 Function 2: LCD test and contrast adjustment

The "**LSP+**" and "**LSP-**" keys can be used to display one of four different test patterns to evaluate the display.

- Pattern 0: displays a portion of the pictograms
- Pattern 1: displays the other portion of pictograms
- Pattern 2: all pictograms and all pixels in text lines are displayed
- Pattern 3: displays patterns 0 and 1 interchangeably (blinking)

The "Light+" and "Light-" keys can be used to incrementally increase or decrease the current contrast adjustment of the display.

2.3.3.3 Function 3: Keyboard and LED test

Keys:

In the text line (after the function number), the keys and bit-specific assignments of keyboard ports are displayed for all three operating panels.

LEDs:

The "**LSP+**" and "**LSP-**" keys can be used to control the status of the different LED groups. (The corresponding bit numbers in the item are displayed on the bottom left as "STATE" and as a hexadecimal number in the field for the position display.)

2.3.3.4 Function 4: LED test and brightness adjustment

In the text line (after the function number), the LEDs and current settings for the LED control register are displayed for all three operating panels.

The "**Light+**" and "**Light-**" keys can be used to change the brightness adjustments of the corresponding operating panel. "Light+" increases the brightness (i.e., the hexadecimal value decreases), and "Light-" decreases the brightness.

2.3.3.5 Very useful Function 5: Switch test

The **"LSP+**" and "**LSP-**" keys can be used to select a switch input to be displayed in plain text.

The respective switch is displayed in the "SWITCH" field on the lower left, and the current status ("HI" or "LO") in the "STATE" field.

The following switches are available in the basic version: **S10**, **S20**, **S21**, **S22**, **S30**, **S31** and **S51**.

The RTT (removable tabletop) version includes the following switches: S10, S20, S21, S22, S30, S31, S33, S40, S41, S42, S43, S44, S45, S46, S47, S48, S49, S50, S51, S60, S61.

The following information (in binary format) is important when performing a remote diagnosis of problems occurring during pallet transfer to/from the trolley. This information should be forwarded to HSC along with a detailed description of the event. If a malfunction is suspected that cannot be explained using information from the TSG or host system (mc file):

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In the text line (after the function number), the “Switches” text is displayed with the bit-specific assignments of important switch ports. Port 9 and 4 of the CPU, as well as port 1 and 0 of the SWT, are displayed. In the last two positions, the current Trolley_State is shown. When configuring a system as non-RTT, the text “No RTT” appears in place of missing values.

The trolley switch text regarding transfer conditions is displayed in the field for the position display below the text line.

Procedure in the event the recommendations in the mc file do not produce results: Change to service mode in case of malfunction (the table cannot be operated). Change to service mode and note the text line as well as the information in the display. Perform a check of the "suspicious" switches by operating "LSP +" and "LSP -". Leave the service mode if no error is present and the table can be operated again. Also refer to: "Debug display".

2.3.3.6 Function 6: Temperature display

The “**LSP+**” and “**LSP-**” keys can be used to display the HW-related temperature measurement configurations, i.e. temperature of screw brake (S23), heat sink temperatures for components A4110 (PWRD) and A4120 (SUPF). This information is provided under the text line (in C, gradient coil temperatures).

In case of defective sensors, the readout will be 0.

As well as the temperatures on the measuring points on the gradient coil.

2.3.3.7 Function 9: End:

After skipping functions 7 and 8 (which are not currently used), the service mode can be exited by pressing the “**Start/Stop Measurement**” key.

2.3.4 Leaving service mode

Press the button **"HDP+"** until the item **"Exit with Start"** appears. Then press **"Start/Stop"**.

2.3.5 Debug display

The debug display allows internal states to be shown on the table display during otherwise unlimited functioning of the table (depending upon the error sequence). In the event of significant errors, the display of these states can aid in investigating error patterns. This information should be forwarded to the HSC along with a detailed description of the event. If a malfunction is suspected that cannot be explained by using information from the TSG or host system (mc file).

Base tables (no Exchangeable Tabletop)

- State number display (lower left in display)

Removable tabletop systems, if the table is in the lift range

- Number display of trolley state ("STATE" flashes)
- Binary display of switch states in the position display

2.3.6 Activating/deactivating the debug display

On one of the control panels, press the buttons **"Air-", "Air+" and "Light-"** simultaneously.

Procedure (if the recommendations according to the mc file do not produce results)

- If possible, switch on the debug display shortly before the error state. The bed remains fully operable. Maintain the last Good state (STATE or Trolley). Produce the error state and maintain the STATE or Trolley STATE.

2.3.7 Debug marking

A special operation generates a pseudo error (not actual error), shown as Message ID 191. If host communication is in effect, this will be recorded in the error log. This allows identification of a situation that should be more closely investigated via log analysis, thus making it easier to find in the information flow. The marked entry then serves as a reference

2.3.8 Initiating debug marking

On one of the control panels, press the buttons **"HDP+" and "LSP+"** simultaneously. The message "Marking current situation as notable by user" is sent to the host. This entry appears in the error log.

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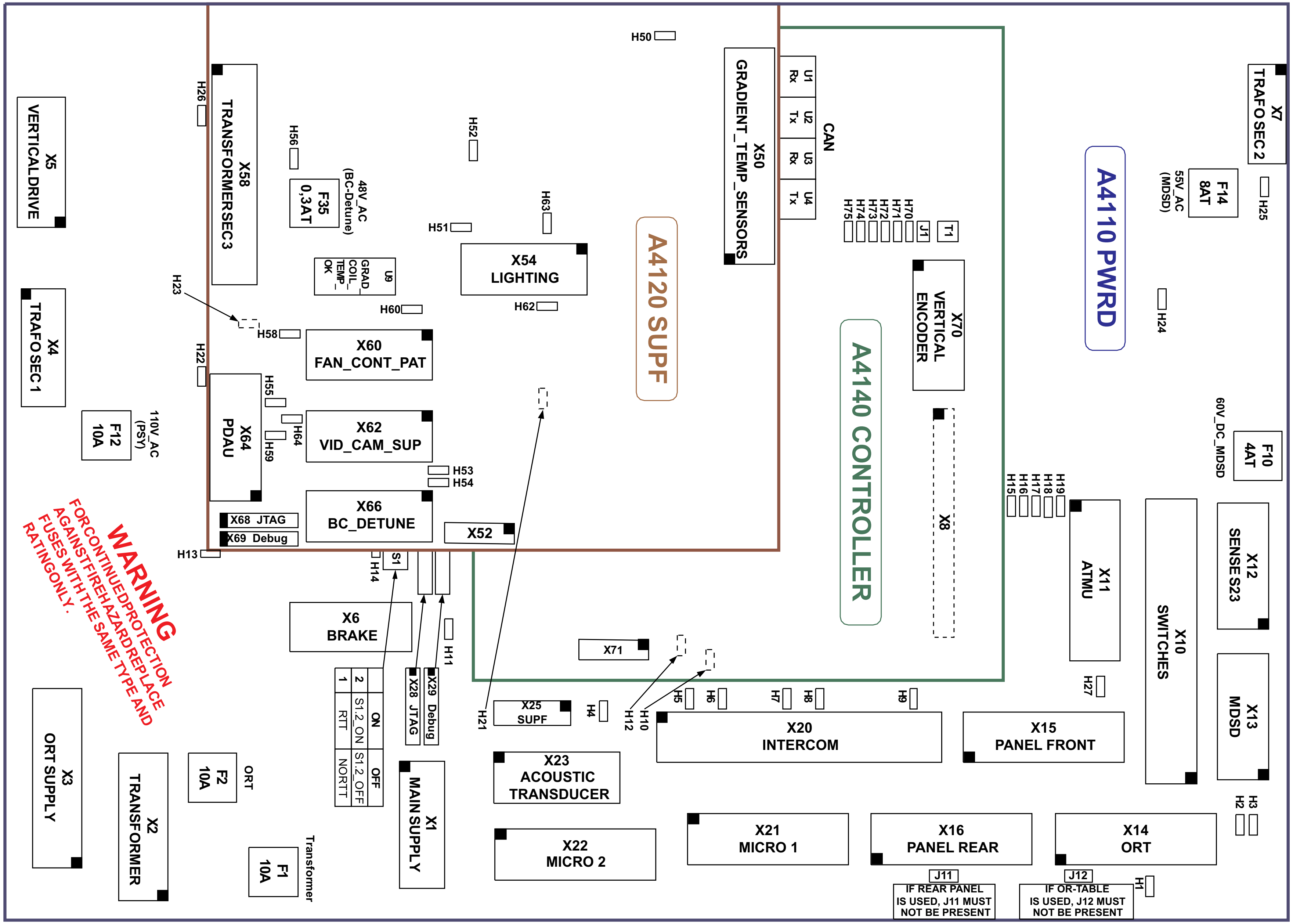
3.1 General

A parts overview of the location of the patient table electronics is depicted on the cover of the patient table electronics. The locations of the fuses, plug connections, PCB's, LEDs are illustrated in this diagram.

Fig. 22: Parts location overview, patient table electronics



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3.2 LEDs

Reference	Color	Name	Function
PWRD - A4110			
H1	Green	+24V_DC_ORT	(conditioned) auxiliary voltage for ORT
H2	Green	+24V_DC_SWT	Switch and sensor power supply (unswitched)
H3	Green	+24V_DC_MDSD	Power supply 24 V MDSD
H4	Green	/WAKE_TMP_COIL	off during wake signal by temperature monitor coils (not used)
H5	Green	/STOP_PAN_R	off with open stop circuit of rear operating panel, J11
H6	Green	/STOP_ICON	off with open stop circuit of intercom
H7	Green	/STOP_PAN_F	off with open stop circuit of front operating panel
H8	Green	/STOP_ORT	off with open stop circuit of ORT, J12
H9	Green	/STOP_SWT	off with open stop circuit of SWT (not used)
H10	Green	/WAKE_PAN_F	off with wake signal from front operating panel
H11	Green	+15V_PWRD	PWRD-internal power supply +15 V
H12	Green	/WAKE_PAN_R	off with wake signal from rear operating panel
H13	Green	+5 Vdc	PWRD-internal power supply +5 V
H14	Yellow	BRAKE	lights when brake vents vertical axis
H15	Green	/WAKE_SWT	off with wake signal by SWT (not used)
H16	Green	+DC_CONT	Power supply controller A4140 (10 V)
H17	Green	+14V_DC_PAN_F	Power supply front control panel
H18	Green	+14V_DC_SWT	Power supply SWT component group (A4115)
H19	Green	+14V_DC_PAN_R	Power supply rear control panel
H21	Green	-5V_PWRD	PWRD-internal power supply -5V

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Reference	Color	Name	Function
H22	Green	+140V_DC_PSY	Power supply network vertical output stage (PWRD)
H23	Green	+15V_H_PWRD	PWRD-internal power supply +15V_H
H24	Green	60V_DC_MDSD	Power supply network horizontal output stage (MDSD)
H25	Green	60V_DC_MDSD_ZK	(switched) Intermediate circuit power voltage horizontal output stage (MDSD) before F10
H26	Green	PSY_DC_BUS	(switched) Intermediate circuit power voltage vertical output stage (PWRD)
H27	Green	+14V_DC_ATMU	Power supply ATMU
SWT - A4115			
H30	Green	+14V_SWT	Power supply SWT component group (A4115)
H31	Green	+5V_SWT	SWT internal power supply 5 V (RTT)
H32	Green	+24V_DC_SWT	Power voltage for switch and sensor S40 (RTT)
H33	Yellow	OUT_DC_SWITCHES	Switched power voltage for sensor S33 (RTT)
H34	Yellow	IR_DATA	Flashes during data reception from the IR remote control unit (RTT)
SUPF - A4120			
H50	Green	+5 V.	SUPF internal power supply +5 V
H51	Green	+15V_SUPF	SUPF internal power supply +15 V
H52	Green	-15V_SUPF	SUPF internal power supply -15 V
H53	Red	ERR_TEMP_GRAD	Temperature rise on gradient coil
H54	Red	ERR_TEMP_BC	not used (temp. rise on res. sensor channels)
H55	Green	13.8V_AC (PDAU Supply)	PDAU power supply
H56	Green	48V_AC (BC detune power supply)	BC detune power supply
H58	Green	FAN_DC_PAT	Equipment fan power supply (approx. 48 V)

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Reference	Color	Name	Function
H59	Green	+24V_PDAU	PDAU power supply
H60	Green	FAN_CONT_PAT	Patient fan control voltage (0-10V)
H62	Green	+DC_LIGHT2	Voltage over Light 2 (TULI)
H63	Green	+DC_LIGHT1	Voltage over Light 1 (TULI)
H64	Green	+13.5V_DC_VID	Video camera power supply +13.5V
CONT - A4140			
H70	Green	+5V_ENC_Y	(switched) Supply voltage for lifting axis encoder
H71	Green	+5V_CONT	Power voltage for controller
H72	Yellow	FW running	Flashes when firmware (boot loader) is running
H73	Green	LW running	Flashes when loadware is running
H74	Red	ERROR1	Error display, error number, see display and host notification
H75	Red	ERROR2	Not used (reserved for additional error display)
PAN - A4x6x			
H1-H3	Red	Rear illumination STOP button	Context-dependent controlled by software
H4-H73	White	Rear illumination for buttons	Context-dependent controlled by software
not visible from outside:			
H80	Green	+14V_DC_PAN	Power voltage
H81	Green	+5V_PAN	PAN internal power supply
MDSD - A4320			

Reference	Color	Name	Function
H90	Green	MDSD_24V	Power voltage
H91	Green	MDSD_60V	(switched) Intermediate circuit power voltage for vertical output stage

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3.3 Circuit breakers

Component group circuit breaker	Value	Network name	LED	Fused units
PWRD - A4110				
F1	10AF*	TRANSFORMER		Transformer A4114
F2	10AF*	ORT		Common line voltage ORT
F3	0.2A	NCR_KEY1		NCR_KEY1
F4	0.2A	NCR_KEY2		NCR_KEY2
F5	0.2A	OUT_DC_SWITCHES	-	Switched power supply for switches;see also F8
F6	0.2A	INTERCOM		(power supply STOP circuit)
F7	1.1A	BRAKE	H14	Brake Y-axis
F8	2.5A	18.6V_AC	H1, H2, H3, H11, H13, H21, H23	AC 18.6V raw voltage for+24V_DC- BRAKE (F18)- Various internal aux. voltages
F9	1.35A	+14V_DC_PAN_R	H19	Power supply rear control panel
F10	4AT*	+60V_DC_MDSD	H25 before F10	Power supply MDSD intermediate circuit
F11	2.5A	+14V_DC_PAN_F	H17	Power supply front control panel
F12	10AF*	110V_AC	H22	Raw voltage for output stage PWRD
F13	0.2A	+14V_DC_SWT	H18	Power supply SWT (A4115)
F14	8AT*	55V_AC	H24	Raw voltage for output stage MDSD
F15	0.5A	+24V_DC_ORT	H1	Conditioned voltage for ORT; (F8)
F16	0.5A	+24V_DC_SWT	H2	Power supply 24 V SWT (A4115), (F8)
F17	0.5A	+24V_DC_MDSD	H3	Power supply 24 V MDSD; (F8)

Component group circuit breaker	Value	Network name	LED	Fused units
F18	9A	11.2V_AC	H16, H17, H18, H19	AC 11.2V raw voltage for: +14V_DC_SWT; +14V_DC_PAN_F; +14V_DC_PAN_R; +DC_CONT BRAKE (---F8)
F19	0.2A	+14V_DC_ATMU	H27	Power supply ATMU
SUPF - A4120				
F30	4A	16.5V_AC	H50, H64	AC 16.5V raw voltage for: Video camera supply +5 V internal SUPF lighting
F31	1.6A	20.6V_AC	H51, H52, H59	AC 20.6V raw voltage for: +24V_DC_PDAU +15V_SUPF -15V_SUPF
F32	2.5AT*	36V_AC	H58	AC 36 V raw voltage for patient fan power supply
F35	0.3AT*	48V_AC	H56	AC 48 V raw voltage for BC detune supply
F36	4A	13.8V_AC	H55	AC 13.8V PDAU power supply
Most circuit breakers are self-resetting once the error has been corrected. It may sometimes be necessary to wait for the PTC to cool, except:* 250 Vac 6.3 x 32 mm fuse following UL 198G				

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Further troubleshooting informations e.g. error codes can be found in the Service Knowledge Base (SKB).

5.1 Switch locations

CAUTION

Magnetic objects may cause injury

Some parts of the patient table consist of magnetic objects!

» **Keep a distance from the magnetic field**

5.1.1 General

This chapter describes the location of the switches and sensors responsible for:

- Reference
- Movement
- Collision detection
- Control

In the summary of switches and sensors, you will find more information about the installation location. All switches and sensors can be identified by numbers (location) added in the table of "Summary of switches and sensors". Please refer to the (➔ Summary of switches and sensors / Page 58)

You will find additional hints for identifying switches and sensors on the patient table, in the circuit diagrams of the "**Vertical Control**" and "**Horizontal Control**". Refer to the **Function Description** “Patient Handling” control electronics.

For more information about the **location of parts** on the patient table, please refer to the **Function Description** Patient Handling > Overview.

5.1.2 Principle of switch adjustment

5.1.2.1 General

The switches are secured by two screws. The switches can be adjusted by loosening the two screws and turning the switch via the rotation point. See (➔ Fig. 23 Page 49).

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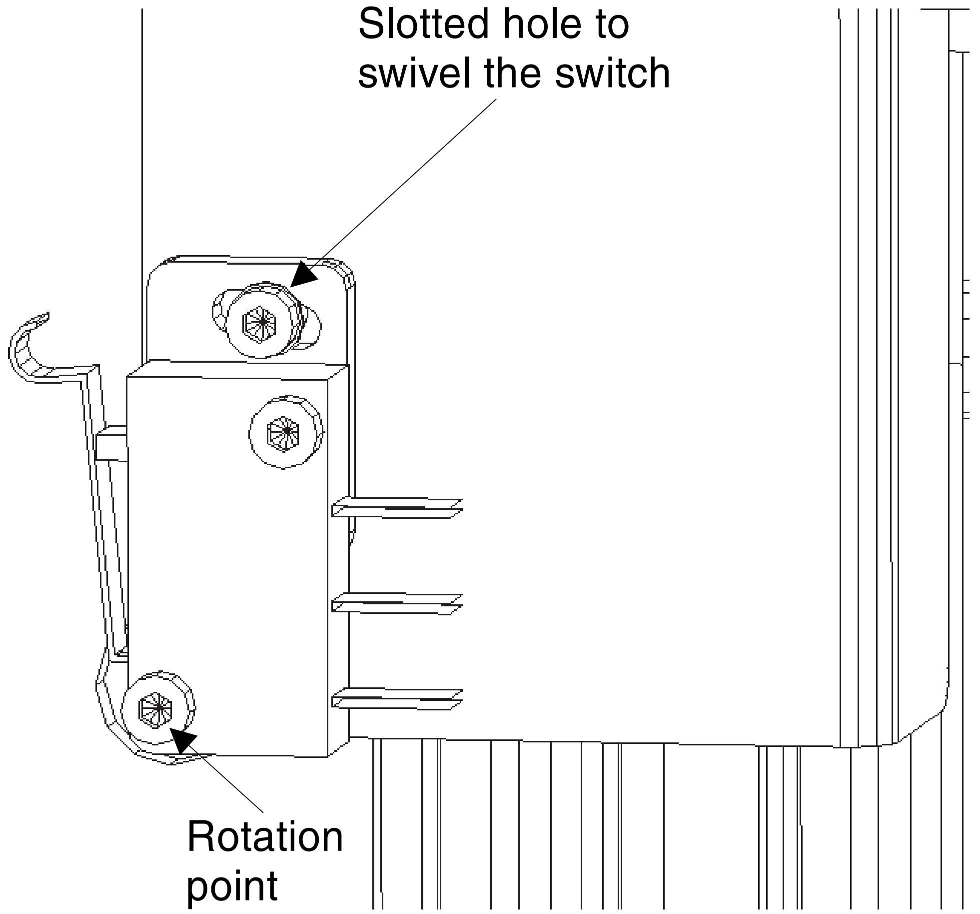
Fig. 23: Princip of switch adjustment



5.1.3 Switches and switch cams on the lifting column

5.1.3.1 General

All switches and sensors on the lifting column can be identified by numbers (location) added in the table of "Summary of switches and sensors". Therefore, the numbers of the switches are part of the switch overview; i.e.:

- (2) S20 Recognition entry height lift
- (3) S21 Advance warning entry height lift, travel upward braking to creep speed

Please refer to the table (→ Summary of switches and sensors / Page 58)

Fig. 24: Switch locations on lifting column

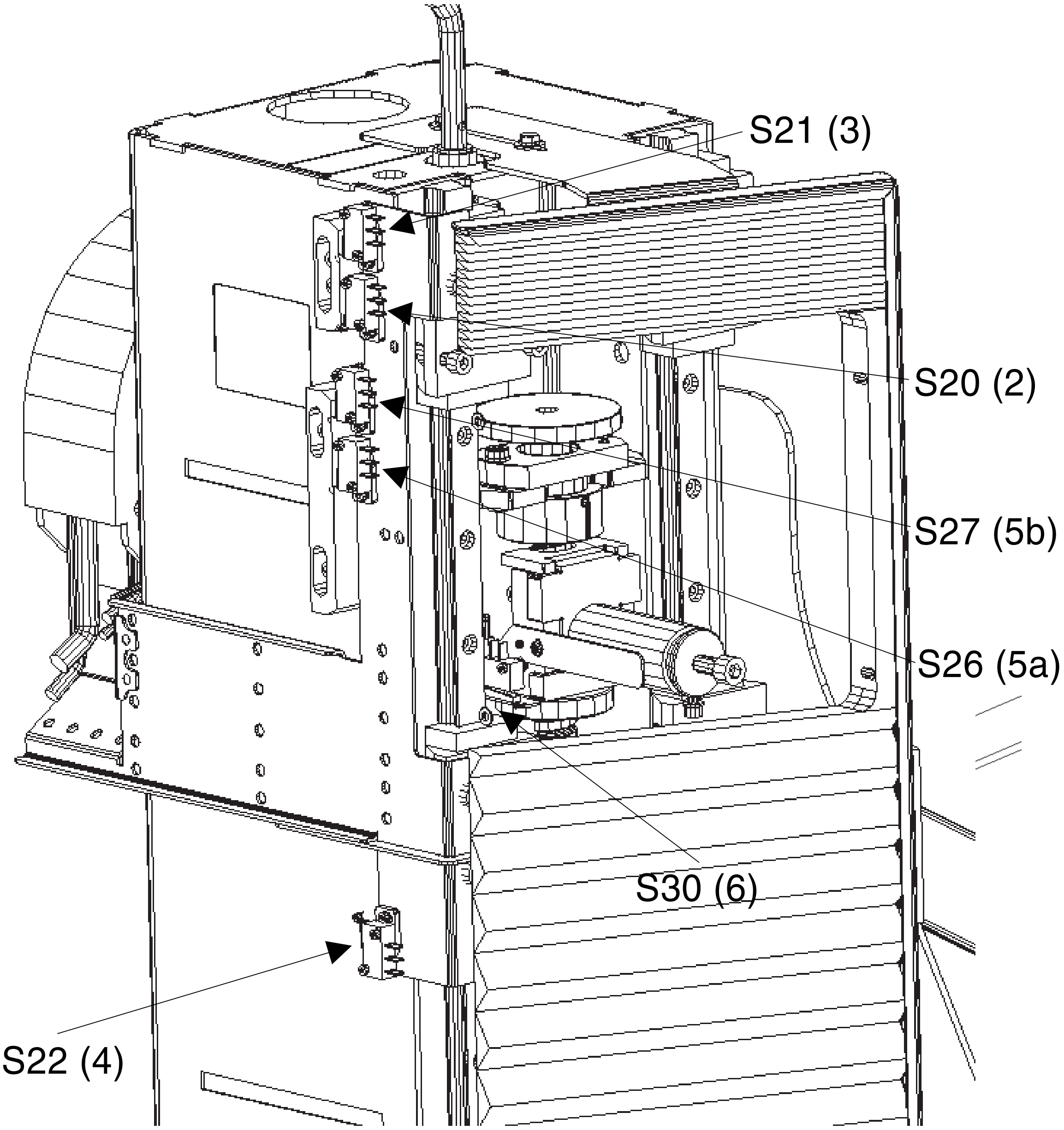


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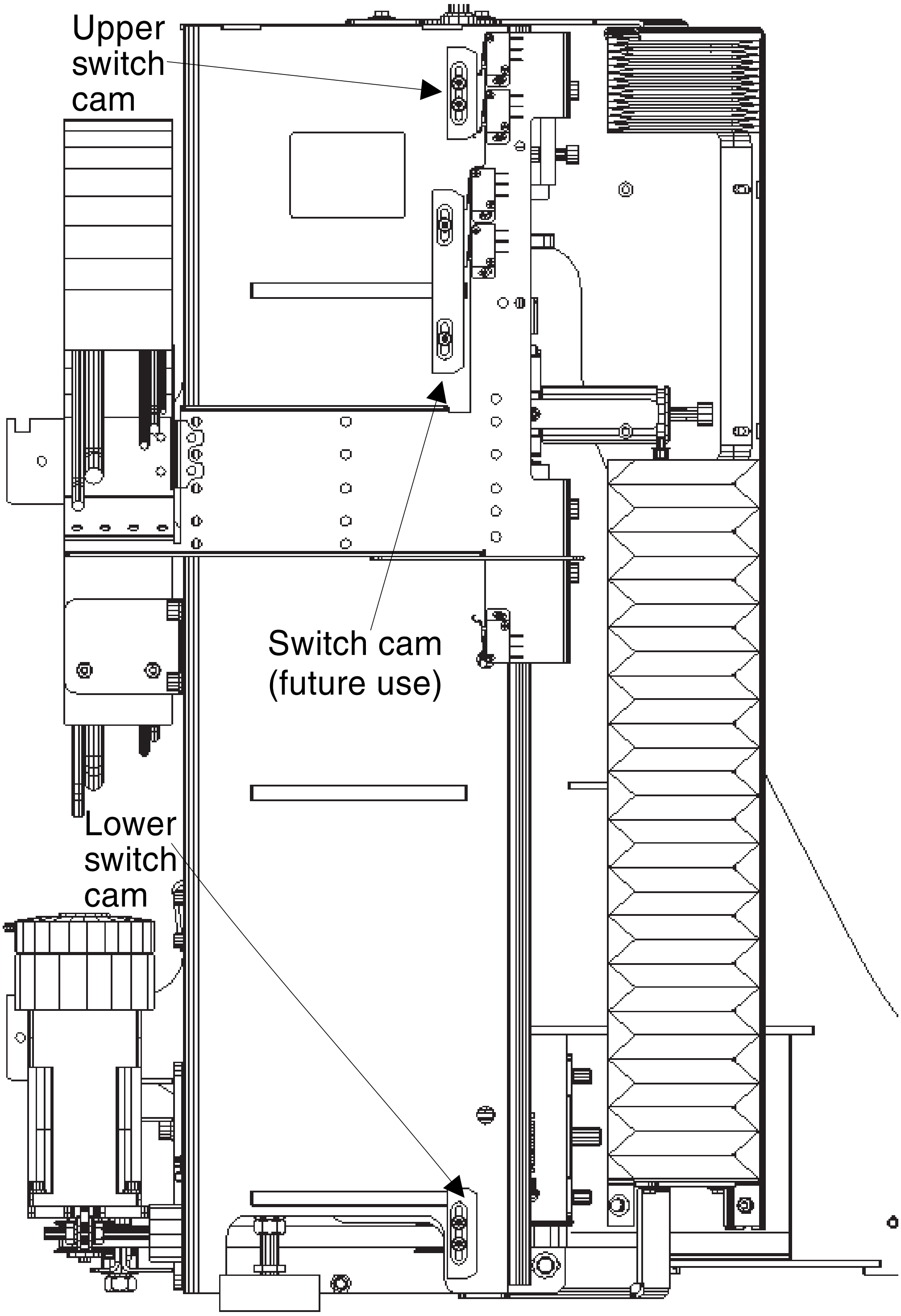
Fig. 25: Switch cams on lifting column



5.1.4 Switches and sensors on the spindle

5.1.4.1 General

All switches and sensors on the spindle can be identified by numbers (location) added in the table of (➔ Summary of switches and sensors / Page 58).

Fig. 26: Princip of upper spindle part

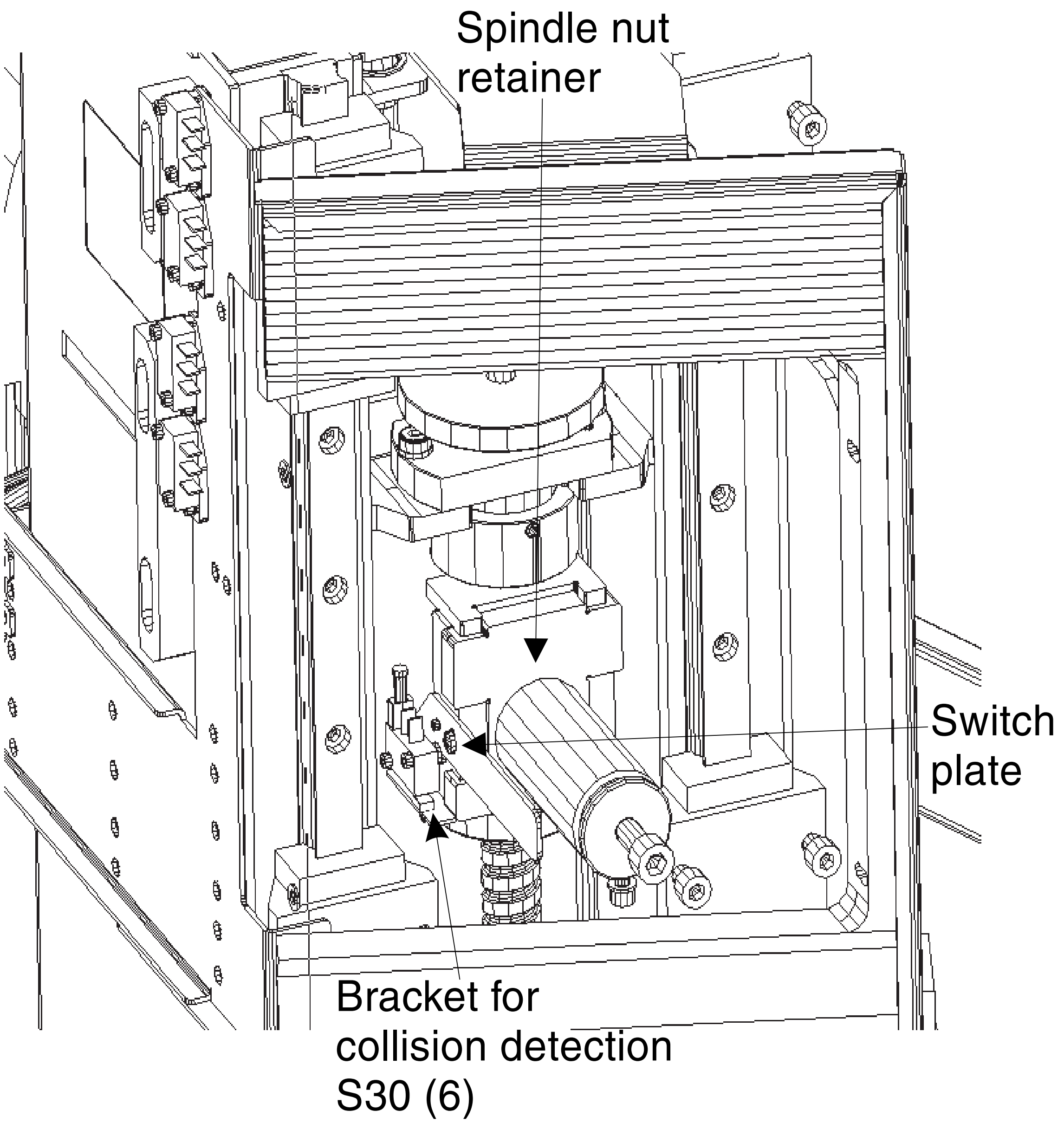


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Fig. 27: Switch location, lower spindle part S23

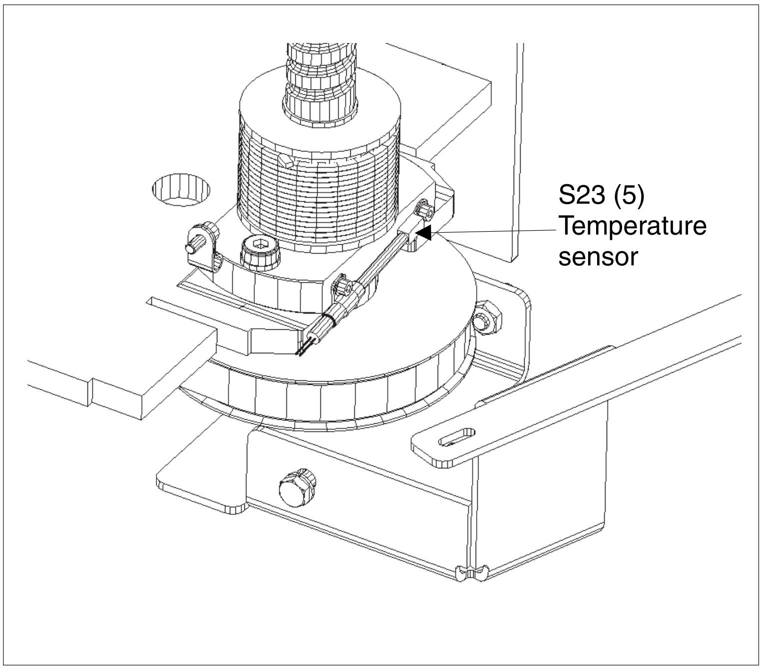
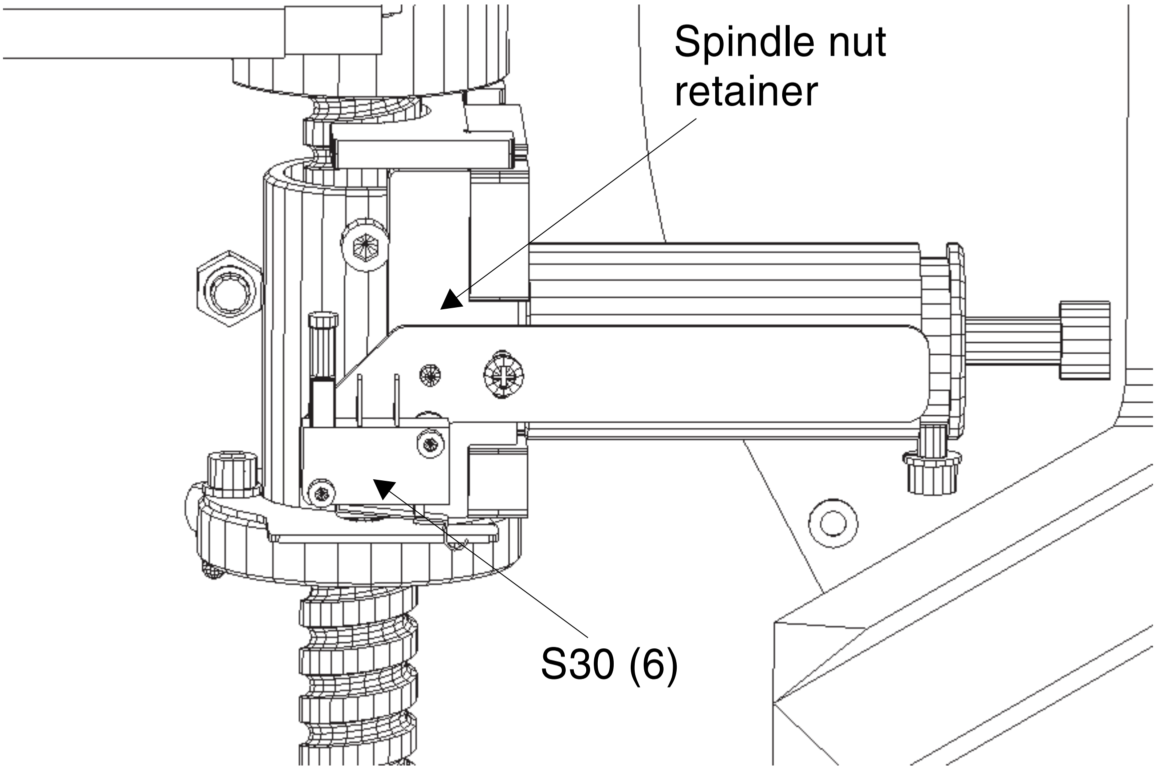


Fig. 28: Switch location upper spindle part



5.1.5 Switches on table top

5.1.5.1 General

All switches and sensors on the spindle can be identified by numbers (location) added in the table of (➔ Summary of switches and sensors / Page 58).

Fig. 29: Switch overview table top

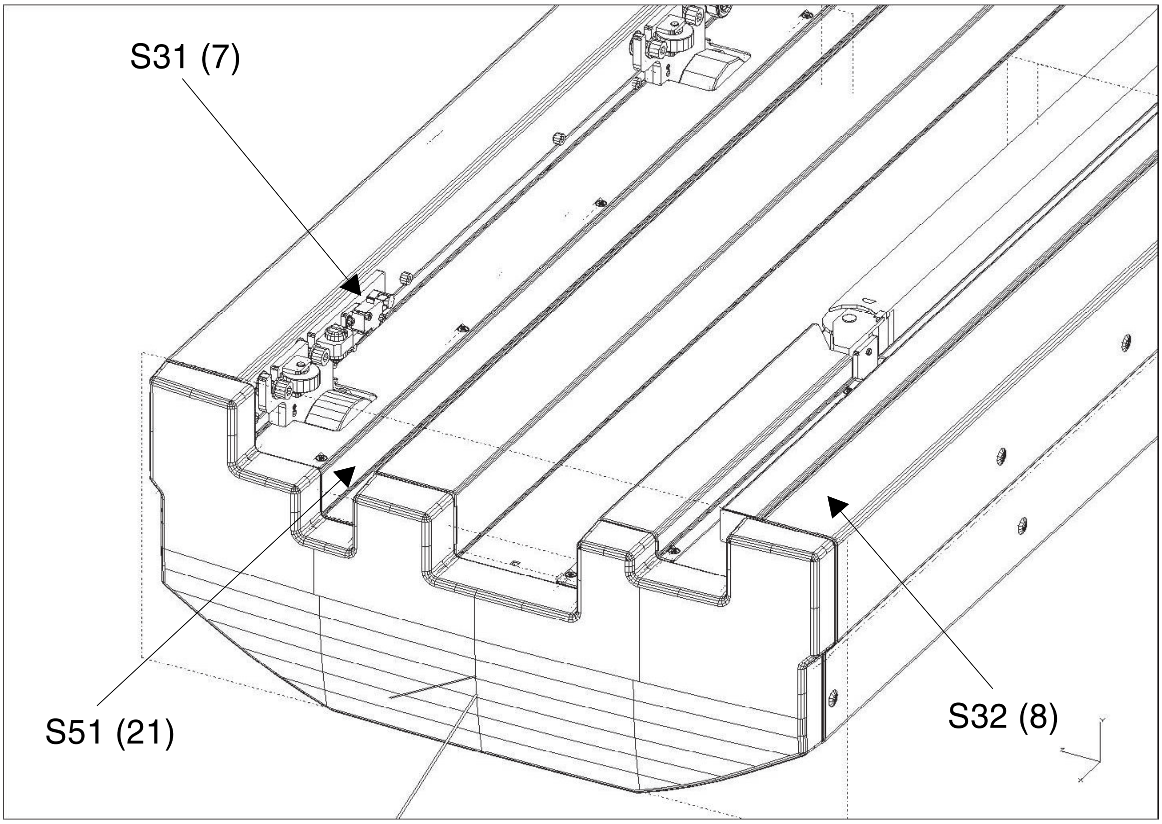


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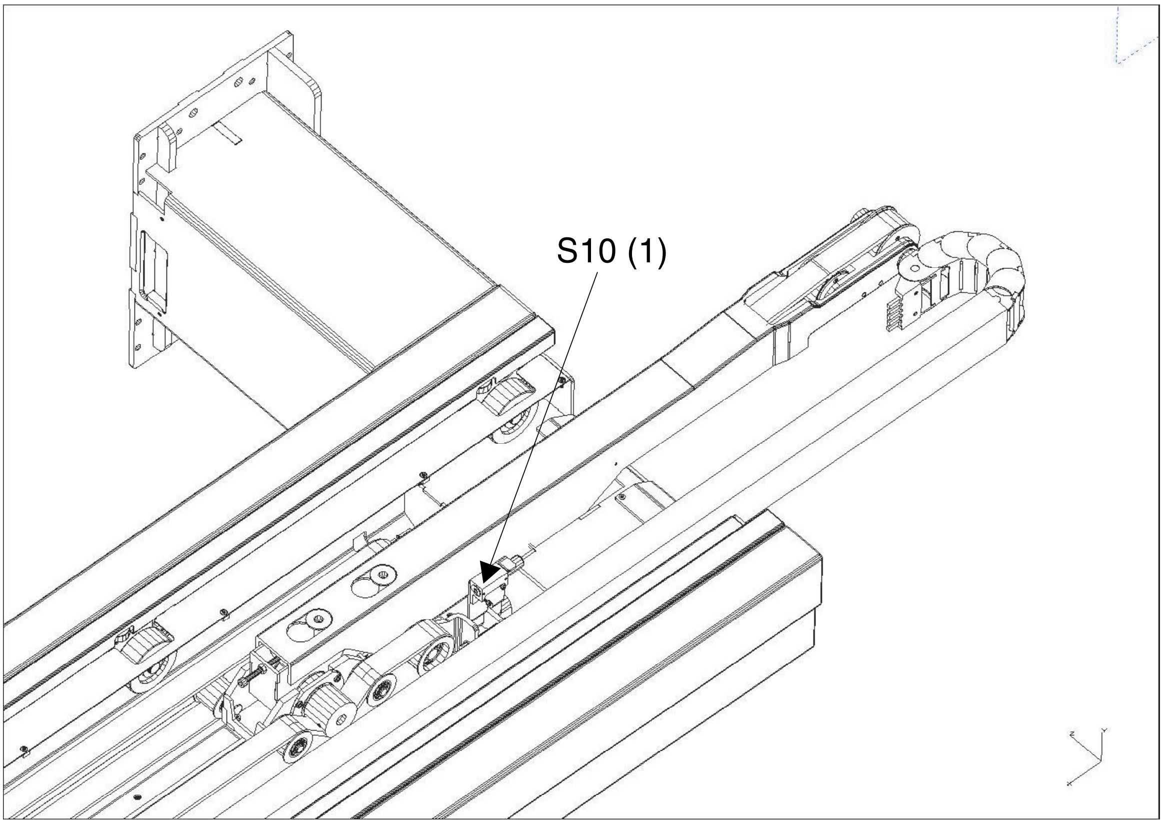
Fig. 30: Reference switch S10



5.1.6 Switches and sensors on the trolley/RTT

5.1.6.1 General

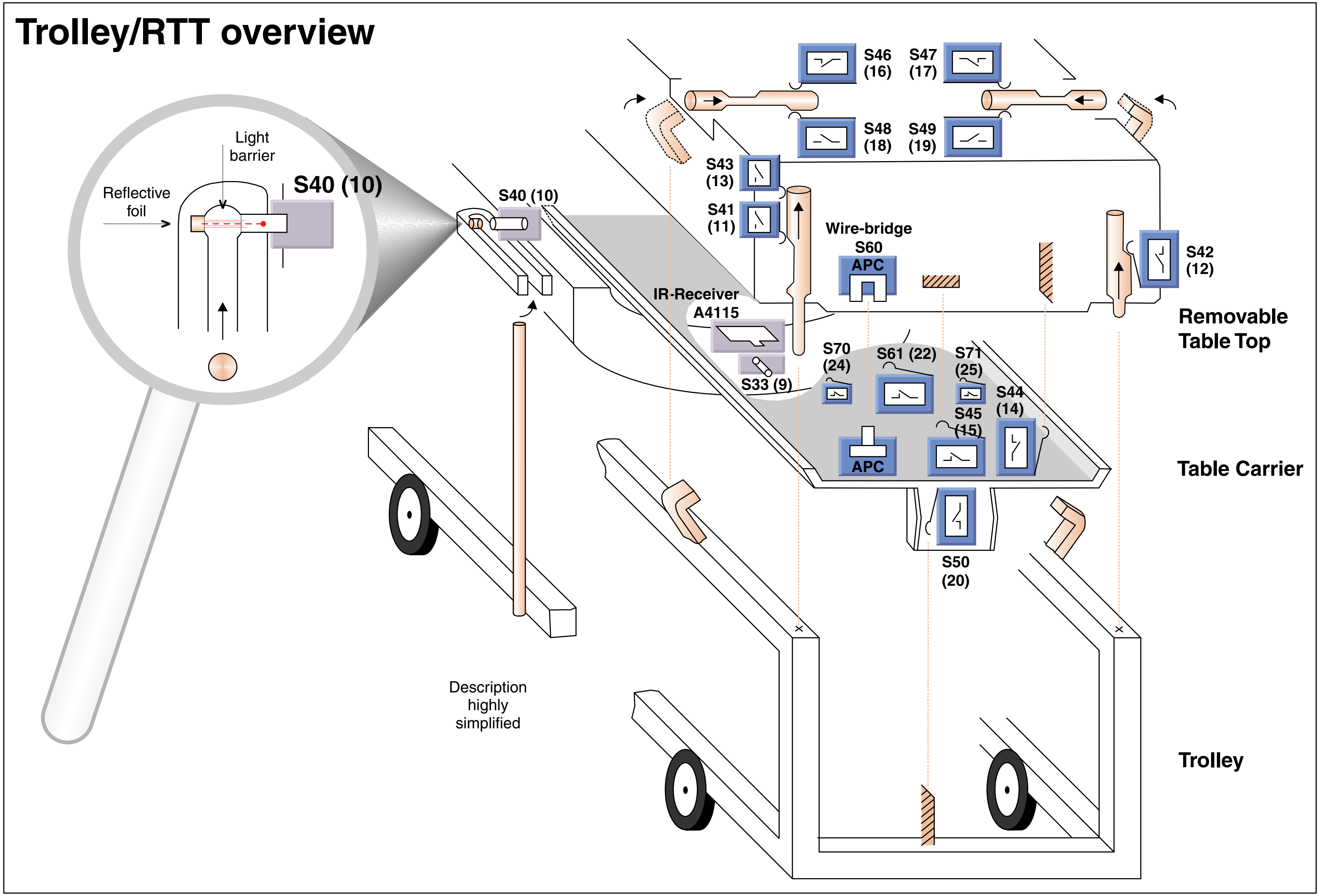
All switches and sensors on the trolley can be identified by numbers (location) added in the table of "Summary of switches and sensors". Therefore, the numbers of the switches are part of the switch overview; i.e.:

- (14) S44 removable tabletop contact during upward travel in approx. 35 mm. (tabletop carrier without removable tabletop below trolley transfer height, removable tabletop on trolley).
- (10) S40 recognition whether the trolley is docked. The trolley guide rod interrupts the sensor's optical connection (reflex photoelectric barrier) with the triple reflector in the docking gripper.

Please refer to the table trolley/RTT overview (➔ Fig. 31 Page 56)

5.1.6.2 Principle of RTT-switches and sensors

Fig. 31: Trolley/RTT Overview



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The multiple switches and sensors perform a plausibility check (as far as possible). In case of illogical combinations, the automatic table stop is triggered and the appropriate error message is displayed. Additional information about the error is contained on the host, i.e., in the error log of the system and mentioned in the MC file. Not all switch errors can be automatically detected.

Some are identifiable through limited functionality.

Behavior observed	Potential defect	Action
No up/down movement, no error message, no direction arrows	- Trolley (or other obstacle) in vicinity is activating S33. - S33 is defective or signal S33_NC is always low	Correctly dock or remove trolley (or obstacle). Check function of S33, refer to Service menu.
Transfer to trolley: Tabletop plate stops correctly at the transfer height, the latches are closed, as indicated by the lock icons, but downward movement is restricted.	S61 is always active or signal S61_NC is always low	Check function of S61, refer to Service menu.
After closing the latches, further downward movement is restricted; the lock icons in the display indicate the latches are open. **A small dot flashes in the shackle of the lock symbol.**	S46 and/or S47 are always activated or signals S46_NC or S47_NC are always low.	Check function of S46 and/or S47, refer to Service menu. Are the APC contacts secured?
After opening the latches, further upward movement is restricted; the lock icons in the display indicate the latches are closed. **A small dot flashes in the shackle of the lock symbol.**	S48 and/or S49 are never activated or signals S48_NC or S49_NC are always low.	Check function of S48 and/or S49, refer to Service menu. Are the APC contacts secured?
Table carrier with removable tabletop is considerably higher than the trolley. However, downwards movement is slow.	S41 is never activated or signal S41_NO is always low	Check function of S41, refer to Service menu.

5.2 Summary of switches and sensors

Switch-name (Location)	RTT only	Functionality	Location	Control state	Condition / Position	Actuating status of switch / sensor	Signal tested	Display in service mode Function 5
S10 (1) (→ Fig. 30 Page 55)		Reference switch horizontal axis (home position). Vertical movement is closed when the table plate is inside the magnet.	Support frame (front-end)	The switch is actuated and the switch contact is closed, if the table plate is fully outside the magnet (home position).	Table plate is fully outside the magnet (home position).	Switch is actuated; switching contact is closed.	S10_NO = high	SWITCH 10 STATE HI
					Table plate is positioned more than approx. 20 mm away from the outer mech. horizontal end stop.	Switch is NOT actuated, switching contact is open.	S10_NO = low	SWITCH 10 STATE LO
S20 (2) (→ Switches and switch cams on the lifting column / Page 49)		Vertical upper end position	Lifting column (Vertical switch assembly)	Switching contact normally closed during vertical movement. It opens during upward movement, shortly before reaching the entry height (0.5 mm).	Lifting axis is at (or above) entry height	Switch is actuated, switching contact is open, horizontal movement is enabled.	S20_NC = low	SWITCH 20 STATE LO
					The lifting axis is more than 0.5 mm, below entry height.	Switch is NOT actuated, switching contact is closed, horizontal movement is blocked.	S20_NC = high	SWITCH 20 STATE HI
S21(3) (→ Switches and switch cams on the lifting column / Page 49)		Vertical slow-speed	Lifting column (Vertical switch assembly)	Opens approx. 40 mm before upper end switch (S20) is reached.	Patient table is in vertical position between upper mech. end stop and 40 mm below the entry height.	Switch is actuated, switching contact is open.	S21_NC = low	SWITCH 21 STATE LO
					Patient table is approx. 40 mm below entry height.	Switch is NOT actuated, switching contact is closed.	S21_NC = high	SWITCH 21 STATE HI

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Switch-name (Loca-tion)	RTT only	Functionality	Loca-tion	Control state	Condition / Position	Actuating status of switch / sensor	Signal tested	Display in service mode Func-tion 5
S22 (4) (➔ Switches and switch cams on the lifting col-umn / Page 49)		Vertical lower end position	Lifting column (Vertical switch assem-bly)	Opens approx. 40 mm before lower mech. end position is reached.	Patient table is in ver-tical position 40 mm above and until reaching the lower mech. end stop.	Switch is actuated, switching contact is open.	S22_NC = low	SWITCH 22 STATE LO
					Patient table is more than 40 mm above the lower mech. stop.	Switch is NOT actu-ated, switching con-tact is closed.	S22_NC = high	SWITCH 22 STATE HI
S23 (5) (➔ Fig. 27 Page 53)		Spindle brake temperature sensor	Lifting column (spindle brake)	Reads actual brake temperature (ana-log sensor).	Brake temperature in-creases during verti-cal down movement. Excessive movement causes brake over-heating.	Normal condition without over-heat-ing: approx. room temp.; the ohmic resistance is approx. 9 - 10 kOhm.	TMP_Y	Tempera-ture display
S24		Reserved						
S26 (5a)		Reserved						
S27 (5b)		Reserved						
S30 (6) (➔ Fig. 26 Page 52)		Vertical colli-sion detection	Lifting column (near spindle nut)	Safety switch, nor-mally actuated. If the table crashes into an obstacle during downwards movement, the support frame loos-ens the contact to the spindle nut and the switch opens.	Patient table has lost contact to the spindle nut (vertical colli-sion).	Switch is NOT actu-ated, switching con-tact is open.	S30_NO = low	SWITCH 30 STATE LO
					Patient table has con-tact to spindle nut = normal state.	Switch is actuated, switching contact is closed.	S30_NO = high	SWITCH 30 STATE HI

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Switch-name (Location)	RTT only	Functionality	Location	Control state	Condition / Position	Actuating status of switch / sensor	Signal tested	Display in service mode Function 5
S31 (7) (→ Fig. 29 Page 54)		Vertical collision detection	Support frame (foot end left)	Safety switch, normally actuated. Tabletop plate vertical collision detection. Switch opens when the table carrier plate is lifted by force. (S31 and S32 are connected in series).	Table carrier plate is lifted by force (vertical collision).	Switch is NOT actuated, switching contact is open.	S31_NC = low (series connection with S32)	SWITCH 31 STATE LO
					Table carrier plate is in normal state.	Switch is actuated, switching contact is closed.	S31_NC = high (series connection with S32)	SWITCH 31 STATE HI
S32 (8) (→ Fig. 29 Page 54)		Vertical collision detection	Support frame (foot end right)	Safety switch, normally actuated. Tabletop plate vertical collision detection. Switch opens when the table carrier plate is lifted by force. (S31 and S32 are connected in series).	Table carrier plate is lifted by force (vertical collision).	Switch is NOT actuated, switching contact is open.	S32_NC = low (series connection with S31)	SWITCH 32 STATE LO
					Table carrier plate is in normal state.	Switch is actuated, switching contact is closed.	S32_NC = high (series connection with S31)	SWITCH 32 STATE HI
S33 (9) (→ Fig. 31 Page 56)	X	Optical sensor for vertical collision protection: Detection of obstacles below the Table carrier plate (e.g. an improperly docked trolley).	Support frame, foot end	Sensor output is low, when an obstacle is detected in the sensor's range (optical reflex sensor). Note: The sensor power voltage is switched off when the trolley is docked correctly (refer to S40).	Obstacle detected within the sensors range or the trolley is docked (then S40 will not be powered). Note: S40 should be checked prior to S33.	Low level of sensor output.	S33_NC = low	SWITCH 33 STATE LO
					No obstacle detected by sensor, no trolley is docked and S40 is powered.	High level of sensor output.	S33_NC = high	SWITCH 33 STATE HI

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Switch-name (Location)	RTT only	Functionality	Location	Control state	Condition / Position	Actuating status of switch / sensor	Signal tested	Display in service mode Function 5
S40 (10) (→ Fig. 31 Page 56)	X	Optical sensor to detect if the trolley is docked. The trolley guide rod interrupts the sensor's light beam to a reflex barrier inside the docking bracket.	Support frame, docking bracket	Sensor output is high, when the trolley is docked.	Trolley guide rod interrupts the sensor's light beam = trolley is docked.	High level of sensor output.	S40 = high	SWITCH 40 STATE HI
					Docking bracket is empty = trolley is NOT docked.	Low level of sensor output.	S40 = low	SWITCH 40 STATE LO
S41 (11) (→ Fig. 31 Page 56)	X	Detection if the RTT is in vicinity to the trolley frame (less the approx. 20 mm).	Removable tabletop (RTT), foot end left side	Switch is not actuated and switching contact is open, if RTT is in vicinity of the trolley frame.	Table carrier plate and RTT are more than 20 mm above the trolley frame. APC is connected.	Switch is actuated, switching contact is closed.	S41_NO = high	SWITCH 41 STATE HI
					Table carrier plate and RTT are lowered down to a position closer than 20 mm to the trolley frame.	Switch is NOT actuated, switching contact is open.	S41_NO = low	SWITCH 41 STATE LO
					If the RTT is separated from the table carrier plate, the APC is not connected.	n.a.	S41_NO = low	SWITCH 41 STATE LO

Switch-name (Location)	RTT only	Functionality	Location	Control state	Condition / Position	Actuating status of switch / sensor	Signal tested	Display in service mode Function 5
S42 (12) (right) (→ Fig. 31 Page 56)	X	Right side of RTT is touching the trolley frame. Contact with trolley in approx. 3 mm. The APC must be fully connected at this position.	Removable tabletop (RTT), foot end right side	Switch is actuated and switching contact is open, when RTT touches the trolley frame (hooks can be closed is this position).	RTT is NOT touching right beam of trolley frame. APC is connected.	Switch is NOT actuated, switching contact is closed.	S42_NC = high	SWITCH 42 STATE HI
					Table is lowered down until RTT touches the right beam of trolley frame.	Switch is actuated, switching contact is open.	S42_NC = low	SWITCH 42 STATE LO
					In case the RTT is separated from the table carrier plate, the APC is not connected.	n.a.	S42_NC = low	SWITCH 42 STATE LO
S43 (13) (left,near APC) (→ Fig. 31 Page 56)	X	Left side of RTT touches the trolley frame. Contact with trolley frame in approx. 2 mm. The APC must be fully connected at this position.	Removable tabletop (RTT), foot end left side	Switch is not actuated and switching contact is open, when RTT touches the trolley frame (hooks can be closed at this position).	RTT is NOT touching left beam of trolley frame. APC is still connected.	Switch is actuated, switching contact is closed.	S43_NO = high	SWITCH 43 STATE HI
					RTT is touching left beam of trolley frame.	Switch is NOT actuated, switching contact is open.	S43_NO = low	SWITCH 43 STATE LO
					In case the RTT is separated from the table carrier plate, the APC is not connected.	n.a.	S43_NO = low	SWITCH 43 STATE LO

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Switch-name (Location)	RTT only	Functionality	Location	Control state	Condition / Position	Actuating status of switch / sensor	Signal tested	Display in service mode Function 5
S44 (14) (➔ Fig. 31 Page 56)	X	Table carrier plate closely before touching the RTT during upward travel (contact in approx. 35 mm.).	Table-carrier plate, foot end	Switch is actuated and switching contact opens closely before the table carrier plate touches the RTT during upward travel.	Table carrier plate and RTT are fully separated.	Switch is NOT actuated, switching contact is closed.	S44_NC = high	SWITCH 44 STATE HI
					Table carrier plate closely before touching the RTT during upward travel (contact in approx. 35 mm.).	Switch is actuated, switching contact is open.	S44_NC = low	SWITCH 44 STATE LO
S45 (15) (➔ Fig. 31 Page 56)	X	Table carrier plate touches RTT fully. At this position the APC has to be fully connected.	Table carrier plate, foot end	Switch is actuated and switching contact is open, when the table carrier plate fully touches the RTT (at this position the hooks can be operated).	Table carrier plate is NOT touching RTT.	Switch is NOT actuated, switching contact is closed.	S45_NC = high	SWITCH 45 STATE HI
					Table carrier plate fully touches RTT.	Switch is actuated, switching contact is open.	S45_NC = low	SWITCH 45 STATE LO
S46 (16) (➔ Fig. 31 Page 56)	X	Left trolley hook assembly is fully closed. At this position the APC has to be fully connected.	RTT foot end left side, close to left hook actuator bolt	Switch is not actuated and switching contact is closed, if left hook is fully closed.	Left trolley hook assembly is fully closed, actuator bolt is fully pressed. RTT is safely closed.	Switch is NOT actuated, switching contact is closed.	S46_NC = high	SWITCH 46 STATE HI
					Left trolley hook assembly is not closed, actuator bolt not fully pressed.	Switch is actuated, switching contact is open.	S46_NC = low	SWITCH 46 STATE LO
					In case the RTT is separated from the table carrier plate, the APC is not connected.	n.a.	S46_NC = low	SWITCH 46 STATE LO

Switch-name (Location)	RTT only	Functionality	Location	Control state	Condition / Position	Actuating status of switch / sensor	Signal tested	Display in service mode Function 5
S47 (17) (→ Fig. 31 Page 56)	X	Right trolley hook assembly is fully closed. At this position the APC has to be fully connected.	RTT foot end right side, close to right hook actuator bolt	Switch is not actuated and switching contact closed, if right hook is fully closed.	Right trolley hook assembly is fully closed, actuator bolt is fully pressed. RTT is safely closed.	Switch is NOT actuated, switching contact is closed.	S47_NC = high	SWITCH 47 STATE HI
					Right trolley hook assembly is not closed, actuator bolt not fully pressed.	Switch is actuated, switching contact is open.	S47_NC = low	SWITCH 47 STATE LO
					In case the RTT is separated from the table carrier plate, the APC is not connected.	n.a.	S47_NC = low	SWITCH 47 STATE LO
S48 (18) (→ Fig. 31 Page 56)	X	Left trolley hook assembly is opened. At this position the APC has to be fully connected.	RTT foot end left side, close to left hook actuator bolt	Switch is actuated and switching contact is closed, if left hook is open.	Left trolley hook assembly is opened.	Switch is actuated, switching contact is closed.	S48_NO = high	SWITCH 48 STATE HI
					Left trolley hook assembly is at least partly closed, actuator bolt at least partly pressed. Note: S46 detects if the actuator bolt is fully pressed and the RTT is safely closed!	Switch is NOT actuated, switching contact is open.	S48_NO = low	SWITCH 48 STATE LO
					In case the RTT is separated from the table carrier plate, the APC is not connected.	n.a.	S48_NO = low	SWITCH 48 STATE LO

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Switch-name (Location)	RTT only	Functionality	Location	Control state	Condition / Position	Actuating status of switch / sensor	Signal tested	Display in service mode Function 5
S49 (19) (➔ Fig. 31 Page 56)	X	Right trolley hook assembly is opened. At this position, the APC has to be fully connected.	RTT foot end right side, close to right hook actuator bolt	Switch is actuated and switching contact is closed, if right hook is open.	Right trolley hook assembly is opened.	Switch is actuated, switching contact is closed.	S49_NO = high	SWITCH 49 STATE HI
					Right trolley hook assembly is at least partly closed, actuator bolt at least partly pressed. Note: S46 detects if the actuator bolt is fully pressed and the RTT is safely closed.	Switch is NOT actuated, switching contact is open.	S49_NO = low	SWITCH 49 STATE LO
					In case the RTT is separated from the table carrier plate, the APC is not connected.	n.a.	S49_NO = low	SWITCH 49 STATE LO
S50 (20) (➔ Fig. 31 Page 56)	X	Vertical collision protection for trolley. End stop for down movement with trolley connected.	Table carrier plate, foot end	Switch is not actuated and switching contact is closed. If the table carrier plate is not in the lowest possible vertical position (shortly before touching the trolley frame).	The table carrier plate is not in the lowest possible vertical position.	Switch is NOT actuated, switching contact is closed.	S50_NC = high	SWITCH 50 STATE HI
					The table carrier plate is at the lowest possible vertical position (shortly before touching the trolley frame).	Switch is actuated, switching contact is open.	S50_NC = low	SWITCH 50 STATE LO

Switch-name (Location)	RTT only	Functionality	Location	Control state	Condition / Position	Actuating status of switch / sensor	Signal tested	Display in service mode Function 5
S51 (21) (→ Fig. 31 Page 56)		Detection if the emergency clutch is operated by the user (motor uncoupled from horizontal drive).	Support frame, foot end at emergency clutch	Switch actuated and switching contact closed (normal operation, motor coupled to horizontal drive).	The emergency clutch is NOT operated by the user; mechanical coupling between motor (MDSD, A4320) and telescopic arm.	Switch is actuated, switching contact is closed.	S51_NO = high	SWITCH 51 STATE HI
					The emergency clutch is operated by the user; no mechanical coupling between motor (MDSD, A4320) and telescopic arm.	Switch is NOT actuated, switching contact is open.	S51_NO = low	SWITCH 51 STATE LO
S52		Reserved						
S60 (23) (→ Fig. 31 Page 56)	X	Detection if APC is connected.	Jumper in APC	Contact closed when APC is properly connected.	APC is connected.	Pins 3-13 of APC socket are bridged by the jumper.	S60 = high	SWITCH 60 STATE HI
					APC is NOT connected.	Pins 3-13 of APC are open.	S60 = low	SWITCH 60 STATE LO

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Switch-name (Location)	RTT only	Functionality	Location	Control state	Condition / Position	Actuating status of switch / sensor	Signal tested	Display in service mode Function 5
S61 (22) (→ Fig. 31 Page 56)	X	Detection if connectors for vacuum cushions, headphones and alarm squeeze ball are disconnected. This is a prerequisite for separation of RTT from the table carrier plate.	Table carrier plate, foot end (close to air hose connectors)	Switch not actuated and switching contact closed, when all air hose connectors are pulled out. Note: S61 is connected in series with S70 and S71.	Air hose connectors are unplugged (and coil plugs X7 and X9 are disconnected).	Switch is NOT actuated, switching contact is closed.	S61_NC = high	SWITCH 61 STATE HI
					One of the connections is plugged.	Switch is actuated, switching contact is open.	S61_NC = low	SWITCH 61 STATE LO
S62	X	Reserved	Jumper in APC					
S63	X	Reserved	Jumper in APC					
S70 (24) (→ Fig. 31 Page 56)	X	Detection of coil plug X7	Table carrier plate, foot end left side, (close to lid of coil plug X7)	Switch is not actuated and switching contact is closed, when lid is closed. No coil plug is plugged into X7. Note: Switch is connected in series to S61 and S71.	Coil plug X7 unplugged.	Lid closed = coil plug X7 unplugged.	S70_NC = high	SWITCH 70 STATE HI
					Coil plug X7 plugged.	Lid open = coil plug X7 unplugged.	S70_NC = low	SWITCH 70 STATE LO

Switch-name (Location)	RTT only	Functionality	Location	Control state	Condition / Position	Actuating status of switch / sensor	Signal tested	Display in service mode Function 5
S71 (25) (→ Fig. 31 Page 56)	X	Detection of coil plug X9	Table carrier plate, foot end right side, (close to lid of coil plug X9)	Switch is not actuated and switching contact is closed, when lid is closed. No coil plug is plugged into X9. Note: Switch is connected in series to S61 and S70.	Coil plug X9 unplugged.	Lid closed = coil plug X9 unplugged.	S71_NC = high	SWITCH 70 STATE HI
					Coil plug X9 plugged.	Lid open = coil plug X9 plugged.	S71_NC = low	SWITCH 71 STATE LO

- RTT only = These switches/sensors are only required to operate a **R**emovable **T**able**T**op system
- APC = Automatic plug connection

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5.3 Switch adjustment

5.3.1 S10 reference switch

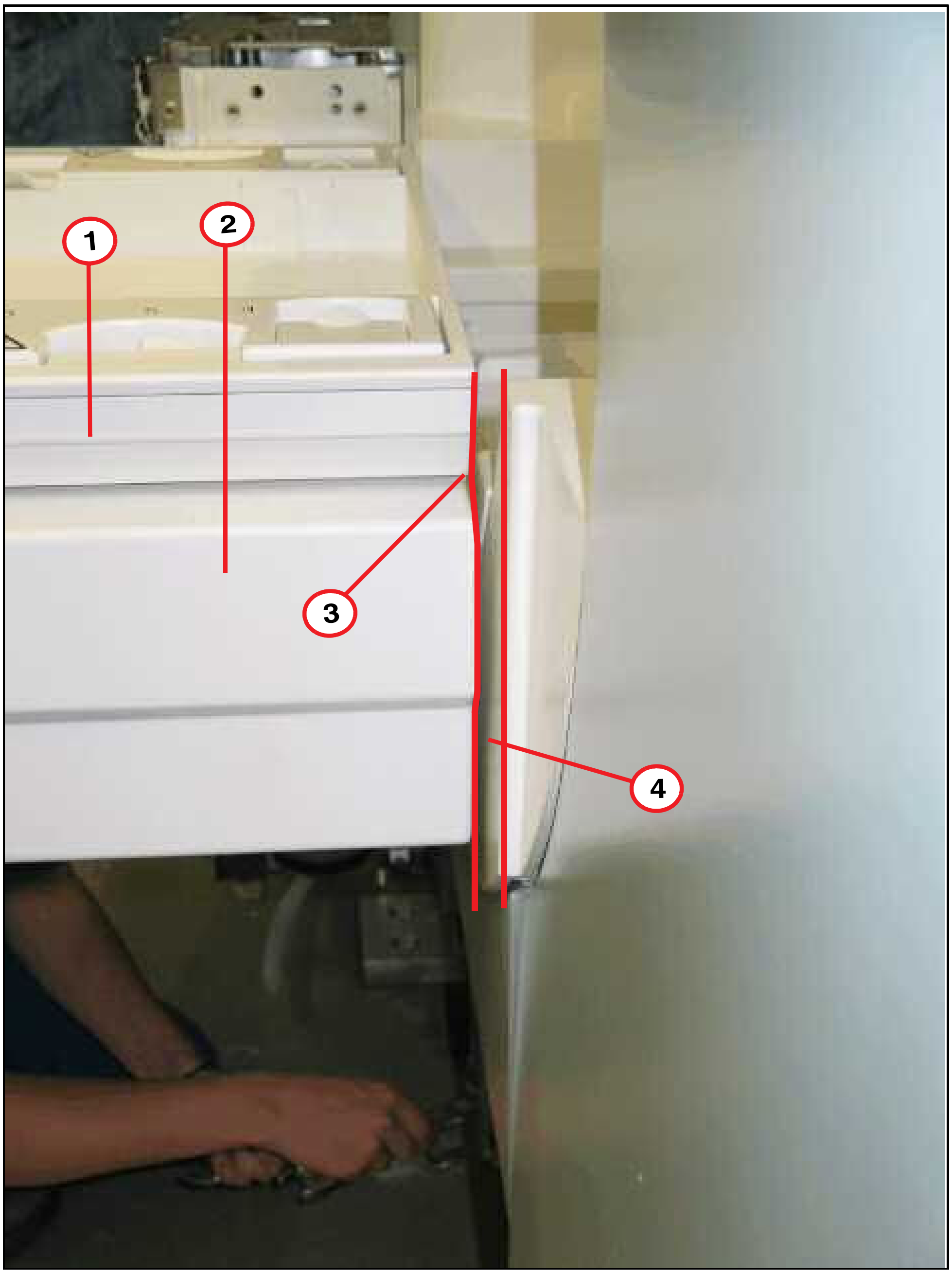
If the reference switch S10 is not properly adjusted, it may happen that the errors **53, 54, 104, 105, 106** occur frequently. For this reason, check S10 according to the following steps:

5.3.2 Switch check

- Visually examine the covers along the seam to the table to ensure they are LEVEL
- Monitor vertical movement(s) at the narrowest position between the table and covers. There should be a clearance of 3 - 5 mm, remain at this position.
- Power off the table, manually (F22) move the tabletop plate to the outer mechanical end stop.
- Power on the table and monitor the subsequent automatic tabletop movement for referencing: During the reference phase, the tabletop plate moves toward the magnet or cover. This should result in the table reference position at the front of the support frame.
- **Check:** The front of the table should be flush with the front of the support frame.

> i The table should NOT touch the cover. Shortly before reaching the cover, the table should move away from the magnet.

Fig. 32: Clearance, reference switch adjustment



(1) Tabletop plate
(2) Tabletop support frame
(3) Adjust switch cam so that the tabletop plate is level with the tabletop support frame
(4) 3 to 5 mm clearance between table and magnet cover

- The tabletop support frame should be adjusted to a clearance of 3 mm to 5 mm between the ptab and the magnet cover.

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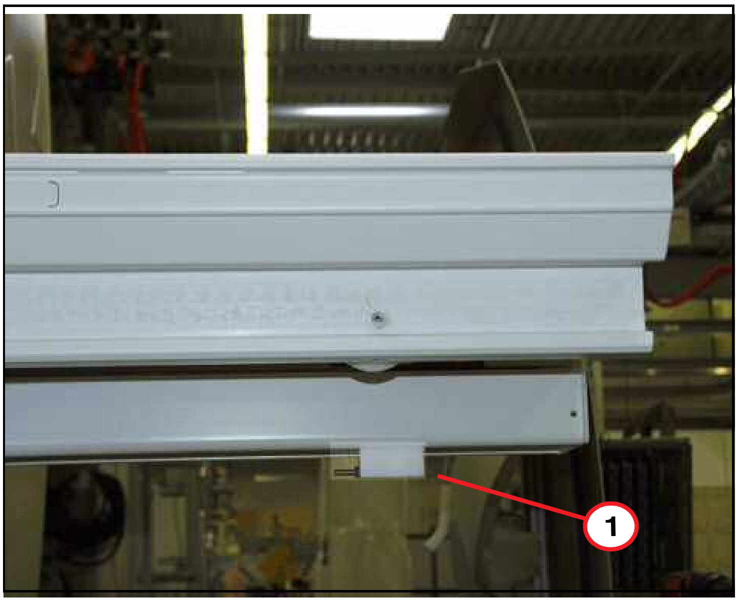
5.3.3 Switch adjustment

If the table top and the support frame are not level: The switch cam, located at the head end of the patient table, needs to be adjusted.

Fig. 33: Switch cam location



(1) Location of S10 switch cam

- For precise setting, the threaded bolt has to be adjusted. This means it has to be screwed into the holder if the table top stops after the support beam, or vice versa. The reference procedure may then be repeated. The front of the table should be flush with the front of the support frame.

After finishing the adjustment procedure tight the counter nut of the switch cam.

Fig. 34: Switch cam S10

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5.4 Switches for configuration

Switch	Label	Function in position ON	Function in position OFF	Default
S1.2 (upper)	S1.2_ON / S1.2_OFF	Reserved	Must be switched to OFF position (norm. stat.)	OFF (right)
S1.1 (lower)	RTT / NO RTT	RTT present. A removable tabletop system is used. The component group SWT must be present in version A4115.	NO RTT present. A removable tabletop system is not used. The version of the component group SWT (A4115 or A4116) is unimportant; A4116 is more useful.	-
The configuration switches are located in control unit A4100 on the component group PWRD A4110, on the left next to X6. The switch should be correctly set by the manufacturer. Modifications in the course of servicing should not be required.				

5.5 Jumpers

Jumper	Installation location	Function when jumper closed	Function when jumper open	Default
J1	CONT A4140	The loadware* stored in the controller is automatically started after power-up. All table functions are available. LED H73 (LW running) should flash (except during sleep mode).	The loadware* stored in the controller is NOT automatically started after power-up. Table functions are NOT available. LED H72 (FW running) should flash. It is possible to start the loadware either by operating T1 or using a corresponding CAN command (by host). If that happens, the table functions are available.	Closed
J12	PWRD A4110,in addition to X14	Bridges the STOP loop (open circuit) of the optional operating table. If a corresponding device is attached to X14, J12 MUST be open.	The bridge of the STOP loop is open; the STOP circuit installed to X14 is active.	Closed
J11	PWRD A4110,in addition to X16	Bridges the STOP loop (open circuit) of the optional "rear panel" control panel (PAN_R). If a control panel is attached to X16, J11 MUST be open; otherwise the STOP button will not function.	The bridge of the STOP loop is open, the STOP circuit installed in X16 is active.	-
firmware:** Non-modifiable component of the software (boot loader) in the controller that is always started after power-up. It enables the loading of new loadware (CAN) as well its initiation. In FW context H72 (yellow) flashes.loadware:** Easily modified software component (table application) in the controller, started depending upon J1, T1 and related CAN commands by the host through the FW. In LW context, H73 (green) flashes. In sleep state, H73 glows continuously. Corresponding CAN commands can switch between LoadWare-LW and Firmware-FW.				

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5.6 Connectors/cables

Sorted by plug connector types (to estimate plug security) **D-Sub's 9female at PWRD, CONT, SUPF**								
	A4140-X70	**A4110-X13**	**A4110-X12**	**A4120-X54**	**A4120-X54**	**A4120-X60**	**A4120-X54**	**A4120-X64**
	Encoder vertical	**MDSD**	**SENS S23**	**TULI**	**TULI**	**FAN PAT**	**VIDEO SUPPLY**	**PDAU SUPPLY**
	D-Sub 9 fm	**D-Sub 9 fm**	**D-Sub 9 fm**	**D-Sub 9 fm**	**D-Sub 9 fm**	**D-Sub 9 fm**	**D-Sub 9 fm**	**D-Sub 9 fm**
Pin	**Signal**	**Signal**	**Signal**	**Signal**	**Signal**	**Signal**	**Signal**	**Signal**
1	+5V_ENC	GND		NTC_6 (1)		GND		-
2	ENC_Y_A	/SIO_Tx_MDS	GND	GND	GND		GND	-
3	ENC_Y_B	MDSD_24V	S23_1	+DC_LIGHT1	+DC_LIGHT1	FAN_SENS_PAT	+13.5V_DC_VID	GND
4	ENC_Y_C	MDSD_24V	S23_2	NTC_6 (2)		FAN_CONT_PAT		13,8V_AC_A
5		GND	GND	GND	GND	GND	GND	13,8V_AC_B
6	GND	M_S10_NO		NTC_7 (1)				+24V_DC_PDAU
7	ENC_Y_/A	SIO_Rx_MDS		+DC_LIGHT2	+DC_LIGHT2			GND
8	ENC_Y_/B	MDSD_DC_BUS		NTC_7 (2)		FAN_DC_PAT		13,8V_AC_A
9	ENC_Y_/N	MDSD_DC_BUS				FAN_DC_PAT		13,8V_AC_B
shell				Valid beginning with Rev. 03	Valid to Rev. 02			

	A4120-X66	LCM119-X1	LCM119-X3	A4410-X101	A4410-X102	A4110-X23	Acoustic transducer
	BC dETUNE	"in"	Light localizer	TULI	TULI	ACOUSTIC	
	D-Sub 9 fm	D-Sub 9 m	D-Sub 9 fm	D-Sub 9 m	D-Sub 9 fm	D-Sub 9 m	D-Sub 9 m
Pin	Signal	Signal	Signal	Signal	Signal	Signal	Signal
1		GND	GND			HP_1	HP_1
2		SCL_PAN	LM	GND	GND	HP_2	HP_2
3	GND	SDA_PAN	LM	+DC_LIGHT1	+DC_LIGHT2	GND	GND
4	48V_AC_A	A0	LM			CALL_SIG	CALL_SIG
5	48V_AC_B	GND	GND	GND	GND	CALL_GND	CALL_GND
6		GND	GND			HP_1	HP_1
7	GND	UB	-	+DC_LIGHT2	+DC_LIGHT1	HP_2	HP_2
8	48V_AC_A	UB	-			CALL_SIG	CALL_SIG
9	48V_AC_B	GND	GND			CALL_GND	CALL_GND
shell		GND	GND	GND		GND	?

PAN, DISP, LM, TULI, ATMU						
	A4110-X11	A4110-X15	A4110-X16	A416x-X80	A416x-X81 A416x-X82	A4110-X14
	AIR TEMP	PANELS FRONT	PANEL REAR	to PWRD	next PAN / DISP	ORT
	D-Sub 15 fm	D-Sub 15 fm	D-Sub 15 fm	D-Sub 15 m	D-Sub 15 fm	D-Sub 15 fm
Pin	Signal	Signal	Signal	Signal	Signal	Signal
1	IIC_E_SCL_ATMU	IIC_E_SCL_PAN_F	IIC_E_SCL_PAN_R	IIC_E_SCL_PAN	IIC_E_SCL_PAN	GND
2	IIC_E_SDA_ATMU	IIC_E_SDA_PAN_F	IIC_E_SDA_PAN_R	IIC_E_SDA_PAN	IIC_E_SDA_PAN	SIO_Rx_ORT

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PAN, DISP, LM, TULI, ATMU						
	A4110-X11	A4110-X15	A4110-X16	A416x-X80	A416x-X81 A416x-X82	A4110-X14
	AIR TEMP	PANELS FRONT	PANEL REAR	to PWRD	next PAN / DISP	ORT
	D-Sub 15 fm	D-Sub 15 fm	D-Sub 15 fm	D-Sub 15 m	D-Sub 15 fm	D-Sub 15 fm
Pin	Signal	Signal	Signal	Signal	Signal	Signal
3	n.c.	/STOP_PAN_F	/STOP_PAN_R	/STOP_PAN	/STOP_PAN	/STOP_ORT
4	+14V_DC_ATMU	+14V_DC_PAN	+14V_DC_PAN	STOP_LOOP_Source	n.c.!	+24V_DC_ORT
5	+14V_DC_ATMU	+14V_DC_PAN	+14V_DC_PAN	+14V_DC_PAN	n.c.!	OUT_ORT1
6	+14V_DC_ATMU	+14V_DC_PAN	+14V_DC_PAN	+14V_DC_PAN	+14V_DC_PAN	OUT_ORT2
7	GND	GND	GND	GND	GND	OUT_ORT3
8	n.c.	NCR_KEY1	NCR_KEY1	NCR_KEY1	NCR_KEY1	GND
9	GND	GND	GND	GND	GND	GND
10	GND	GND	GND	GND	GND	/SIO_Tx_ORT
11	n.c.	/WAKE_PAN	/WAKE_PAN	/WAKE_PAN	/WAKE_PAN	OUT_ORT4
12	n.c.	n.c.	n.c.	A1	A1	IN_ORT5
13	n.c.	n.c.	n.c.	A0	A0	IN_ORT6
14	GND	GND	GND	GND	GND	GND
15	n.c.	NCR_KEY2	NCR_KEY2	NCR_KEY2	NCR_KEY2	/ORT_PRESENT
shell	GND	GND	GND	GND	GND	GND

MICROP1, MICRO2		
	A4110-X21	A4110-X22
	MICRO1	MICRO2
	D-Sub 15 m	D-Sub 15 m
Pin	Signal	Signal
1	MIC_SIG_1	
2	MIC_SHIELD_1	
3	MIC_SHIELD_0	
4	MIC_VCC	
5		MIC_VCC
6		MIC_SHIELD_0
7		MIC_SHIELD_2
8		MIC_SIG_2
9	MIC_GND_1	
10		
11	MIC_GND_0	
12		
13		MIC_GND_0
14		
15		MIC_GND_2
shell	GND	GND

Switches, Sensors, Intercom, SUPF				
	A4120-X50	A4115-X31 (SWT) W4142-X31		W4142-APC
	Tempera-ture, gradi-ent coil	Switches at removable tabletop (via APC)		
	D-Sub 25 m	D-Sub 25 fm		APC
Pin	Signal	Pin	Signal	Pin
1		1	GND	8
2		2	S64	2

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Switches, Sensors, Intercom, SUPF				
	A4120-X50	A4115-X31 (SWT) W4142-X31		W4142-APC
	Temperature, gradient coil	Switches at removable tabletop (via APC)		
	D-Sub 25 m	D-Sub 25 fm		APC
Pin	Signal	Pin	Signal	Pin
3	GND	3	+24V_DC_SWITCHES (S64)	3
4		4	S60	1
5		5	+24V_DC_SWITCHES (S60)	13
6		6	S49_NC	23
7		7	+24V_DC_SWITCHES (S47, S49)	22
8	NTC_5 (2)	8	S48_NC	5
9	NTC_4 (2)	9	+24V_DC_SWITCHES (S46, S48)	4
10	NTC_3 (2)	10	S47_NO	21
11	NTC_2 (2)	11	+24V_DC_SWITCHES (S41, S43)	16
12	NTC_1 (2)	12	S46_NO	7
13	NTC_0 (2)	13	GND	6
14		14	GND	18
15	GND	15	OUT_DC_SWITCHES	24
16	GND	16	GND	-
17		17	+24V_DC_SWITCHES (S63)	11
18		18	S63_NC	12
19		19	+24V_DC_SWITCHES (S62)	15
20	NTC_5 (1)	20	S62_NC	14
21	NTC_4 (1)	21	+24V_DC_SWITCHES (S42)	9
22	NTC_3 (1)	22	S43_NO	19

Switches, Sensors, Intercom, SUPF				
	A4120-X50	A4115-X31 (SWT) W4142-X31		W4142-APC
	Tempera-ture, gradi-ent coil	Switches at removable tabletop (via APC)		
	D-Sub 25 m	D-Sub 25 fm		APC
Pin	Signal	Pin	Signal	Pin
23	NTC_2 (1)	23	S42_NC	10
24	NTC_1 (1)	24	S41_NO	17
25	NTC_0 (1)	25	GND	20
shell	GND	shell		

	A4110-X10 (PWRD) (D-Sub 37 fm) A4115-X30 (SWT) (D-Sub 37 m)	A4110-X20 (PWRD) (D-Sub 37 m)
	Switches	Intercom
	D-Sub 37	D-Sub 37
Pin	Signal	Signal
1	GND	n.c.
2	IIC_E_SCL_SWT	GND
3	S41_NO	POT_LSP
4	S42_NC	POT_HP
5	S43_NO	MIC_SIG_1
6	S45_NC	MIC_GND_1
7	S62_NC	MIC_VCC
8	S63_NC	MIC_GND_0
9	S10_NO	CALL_SIG
10	+24V_DC_SWT	CALL_GND
11	GND	HP_1
12	OUT_DC_SWITCHES	HP_2
13	S21_NC	NCR_KEY1
14	S22_NC	NCR_KEY2
15	S26_NC	MIC_SIG_2

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	A4110-X10 (PWRD) (D-Sub 37 fm) A4115-X30 (SWT) (D-Sub 37 m)	A4110-X20 (PWRD) (D-Sub 37 m)
	Switches	Intercom
	D-Sub 37	D-Sub 37
Pin	Signal	Signal
16	S27_NC	MIC_GND_2
17	GND	/STOP_ICOM
18		
19		+24V_DC_INTERCOM
20	GND	GND
21	IIC_E_SDA_SWT	
22	GND	GND
23	S44_NC	GND
24	+14V_SWT	MIC_SHIELD_1
25	/STOP_SWT	
26	/WAKE_SWT	MIC_SHIELD_0
27	GND	
28	S24_NO	GND
29	+24V_DC_SWT	
30	S30_NO	GND
31	S31_NC	
32	S20_NC	GND
33	S50_NC	
34	S51_NC	MIC_SHIELD_2
35	GND	
36		GND
37		
shell	GND	GND

Molex method of counting (ATTENTION: varies from the imprinting on the plugs, but motherboard labeling is correct):

(Pin 1 marked)

1	3	5	...	13
2	4	6	...	14

	A4115-X33 (SWT) W4143-X33 (cable)	W4143-Leads on switch	A4115-X35 (SWT) A4115-X36 (spare) W4144-X35 (cable)
	Switches		Switches
	SWT: Molex: 5566, (14 Pin) 12 pins populated Cable: Molex:5557, 12 Pin		SWT: Molex: 5566, 10 Cable: Molex: 5557, 10 Pin
Pin	Signal	Signal	Signal
1	GND	Shield	GND
2	+24V_DC_SWT	S20_COM (1)	+24V_DC_SWT
3	S20_NC	S20_NC (2)	S50_NC
4	S21_NC	S21_NC (2)	S44_NC
5	S22_NC	S22_NC (2)	+24V_DC_SWT
6	+24V_DC_SWT	S21_COM (1)	+24V_DC_SWT
7	+24V_DC_SWT	S22_COM (1)	S45_NC
8	GND		S61_NC
9	S26_NC	S26_NC (2)	+24V_DC_SWT
10	+24V_DC_SWT	S26_COM (1)	GND
11	S27_NC	S27_NC (2)	
12	+24V_DC_SWT	S27_COM (1)	
13	(S24_NO)		
14	+24V_DC_SWT		

	W4144-Leads on switch	A4115-X34 W4151-X34	W4151-Leads on switch
	Switches	Switches	Switches
		SWT: Molex: 5566, 10 Cable: Molex: 5557, 10 Pin	
Pin	Signal	Signal	Signal
1	Shield	GND	Shield

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	W4144-Leads on switch	A4115-X34 W4151-X34	W4151-Leads on switch
	Switches	Switches	Switches
		SWT: Molex: 5566, 10 Cable: Molex: 5557, 10 Pin	
Pin	Signal	Signal	Signal
2	S44_COM (1)	+24V_DC_SWT	
3			
4	S44_NC (2)	S52	
5		+24V_DC_SWT	S51_COM (1)
6	S61_COM (1)	+24V_DC_SWT	
7	S45_NC (2)	S51_NO	S51_NO (3)
8	S61_NC (2)	S24_NO	
9	S45_COM (1)	+24V_DC_SWT	
10		GND	
11			
12			
13			
14			

	A4115-X37 (SWT) W4145-X37 (Cable)	W4145-Leads on switch	A4115-X38 (SWT) W4146-X38 (Cable)
	Switch S10		Switch S30
	SWT: Molex: 5566, 4 Pin Cable: Molex:5557, 4 Pin		SWT: Molex: 5566, 4 Pin Cable: Molex:5557, 4 Pin
Pin	Signal	Signal	Signal
1	GND	Shield	GND
2	+24V_DC_SWT	S10_COM (1)	+24V_DC_SWT
3	S10_NO	S10_NO (4)	S30_NO
4	GND		GND

	W4146-Leads on switch	A4115-X43 (SWT)	A4115-X39 (SWT) W4147-X39 (Cable)
		spare for Switch S50	Switch S31
		SWT: Molex: 5566, 4 Pin Cable: Molex:5557, 4 Pin	SWT: Molex: 5566, 4 Pin Cable: Molex:5557, 4 Pin
Pin	Signal	Signal	Signal
1	Shield	GND	GND
2	S30_COM (1)	+24V_DC_SWT	S32_NC
3	S30_NO (2)	S50_NC	S31_NC
4		GND	GND

	W4147-Leads on switch	A4115-X40 (SWT) W4148-X40 (Cable)	W4148-Leads on switch
		Switch S32	
		SWT: Molex: 5566, 4 Pin Cable: Molex:5557, 4 Pin	
Pin	Signal	Signal	Signal
1	Shield	GND	Shield
2	S31_COM (1)	+24V_DC_SWT	S32_COM (1)
3	S31_NC (2)	S32_NC	S32_NC (2)
4		GND	

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	A4115-X41 (SWT) A4115-X42 (SWT)	W4149-X41 (Cable)	W4149- S33	W4150-X42 (Cable)	W4150- S40
	Sensors S33, S40	Sensor S33		Sensor S40	
	Molex: 5566, 6 Pin	Molex: 5557, 6 Pin	Sensor plug 4 Pin Wenglor TC66PA3	Molex: 5557, 6 Pin	Sensor plug 4 Pin Wenglor RO88PD3
Pin	Signal	Signal	Signal	Signal	Signal
1	GND	Shield	Shield	Shield	Shield
2	+24V_DC_SWT			+24V_DC_SWT	Pin 1 (+)
3	S33_NC	S33_NC	Pin 2 (OUT)		
4	S40			S40	Pin 2 (OUT)
5	OUT_DC_SWITCHES	OUT_DC_SWITCHES	Pin 1 (+)		
6	GND		Pin 3 (-)		Pin 3 (-)

	A4115-X44 (SWT)
	spare
	SWT: Molex: 5566, 4 Pin Cable: Molex:5557, 4 Pin
Pin	Signal
1	GND
2	+24V_DC_SWT
3	/STOP_SWT
4	GND

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Power pack			
	A4110-X1 (input)	A4110-X3 (opt. LOC)	A4110-X2 (Power pack -pri)
	MAIN SUPPLY	LOC	Power pack
	Wago 4-pin male	Wago 4-pin female	Wago 3-pin female
Pin	Signal	Signal	Signal
1	N (230 Vac)	N (230 Vac)	N (230 Vac)
2	L (230 Vac)	L (230 Vac)	PE - shielding
3	PE	PE	L (230 Vac)
4	Shield	Shield	

	A4110-X4			A4110-X7		
	Power pack sec I			Power pack sec II		
	Wago 5-pin male			Wago 4-pin male		
Pin	Signal		Used for	Signal		Used for
1	18.6V_AC_B	W9	Switches, sensors, aux. voltage, MDSD	11.2V_AC_B	W12	Logic 5 Vdc, controls, dis-plays
2	18.6V_AC_A	W9		11.2V_AC_A	W12	
3	Shield	W8		55V_AC_B	W10	Horizontal operation
4	110V_AC_B	W11	Vertical operation	55V_AC_A	W10	
5	110V_AC_A	W11				

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	A4120-X58		
	Power pack sec. III		
	Wago 10 pin male		
Pin	Signal		Used for
1	36V_AC_A	W4	Fan intermediate circuit
2	36V_AC_B	W4	
3	20.6V_AC_A	W7	PDAU, Temp. mon.
4	20.6V_AC_B	W7	
5	16.5V_AC_A	W5	Light intermed. circuit, video, logic 5 Vdc, temp. mon.
6	16.5V_AC_B	W5	
7	13,8V_AC_A	W6	PDAU display
8	13,8V_AC_B	W6	
9	48V_AC_A	W3	BC detune
10	48V_AC_B	W3	

Vertical drive			
	A4110-X5		A4110-X6
	Vertical drive		BRAKE
	Wago 6 pin		Wago 3-pin
Pin	Signal	JAT normal color	Signal
1	GND (PE)	gn/ge	GND (spare: Shield)
2	/B	bn	BR+
3	B	rt	BR-
4	/A	or	
5	A	sw	
6	GND (Shield)		

Internal Connections				
	A4110-X25	A4120-X51	A4120-X52	A4120-X57
	SUPF	spare	PWRD	PWRD
	Molex 10 m	Molex 10 m	Molex 10 m	Molex 10 m
Pin	Signal	Signal	Signal	Signal
1	GND	GND	GND	GND
2	POT_LSP	nc	POT_LSP	POT_LSP
3	+DC_CONT	nc	nc	nc
4	POT_HP	nc	POT_HP	POT_HP
5	IIC_I_SDA	IIC_I_SDA	IIC_I_SDA	nc
6	GND	GND	GND	GND
7	IIC_I_SCL	IIC_I_SCL	IIC_I_SCL	nc
8	GND	GND	GND	GND
9	/WAKE_TMP_COIL	/WAKE_TMP_COIL	/WAKE_TMP_COIL	nc
10	GND	GND	GND	GND

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6.1 General

This section of the TSG deals with problems regarding the video monitoring. The strategies and procedures provide the necessary information for servicing the video monitoring.

The primary components of the patient video monitoring are:

Patient CCD camera

Patient Video Display

CAUTION

Strong magnetic field.

Magnetizable tools or objects introduced into the magnetic field become dangerous projectiles.

» **Use only non-magnetizable tools and do not carry magnetizable objects when working within the magnetic field. Use utmost caution to avoid hazardous conditions with regard to equipment and personnel.**

6.2 Additionally required documents

Function Description "Patient Handling" chapter

Replacement of Parts "Patient Handling" chapter

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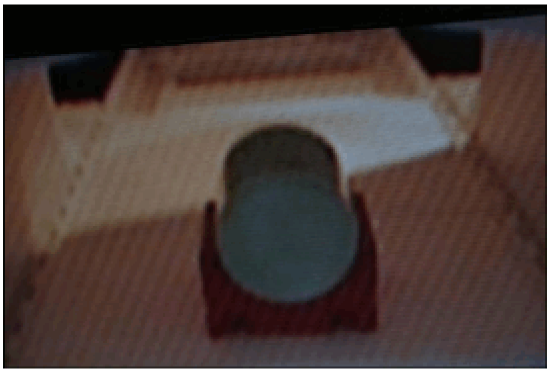
6.3 Common problems

Moire structure or lines with different intensity, frequency and orientation on the patient video display:

Due to the magnetic field, the CCD camera may generate a moire structure or lines on the patient video display (→ Fig. 35 Page 91). To check that the CCD camera is causing the moire structure or lines on the patient video display, remove at the rear of the magnet the upper center camera cover and dismount the camera. Move the CCD camera approx. 30cm away from the magnet and check if the moire structure on the patient video display is gone (a second person may be needed). In case the moire structure is gone after positioning the camera 30cm away from the high field area, only a replacement of the CCD camera will help. A very low intensity moire structure, barely visible, does not indicate a problem, if this is the case the CCD camera should not be replaced.

Fig. 35: Strong moire structure on patient video display



- Advanced trouble shooting procedure for VCG triggering added.

Version 06

Error massage: Linked to SKB.

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